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Paleomagnetic constraints on the paleolatitude of the Lhasa block during the Early Cretaceous: Implications for the onset of India–Asia collision and latitudinal shortening estimates across Tibet and stable Asia

Zhenyu Li [a,*], Lin Ding [a,b], Peiping Song [a], Jiajun Fu [a], Yahui Yue [a]

[a] Key Laboratory of Continental Collision and Plateau Uplift, Institute of Tibetan Plateau Research, Chinese Academy of Sciences, Beijing 100101, China
[b] Center for Excellence in Tibetan Plateau Earth Sciences, Chinese Academy of Sciences, Beijing 100101, China

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ABSTRACT

Interbedded volcano-sedimentary sequences are well exposed in the northern part of the Lhasa block in southern Tibet. Zircon U–Pb dating results from two samples indicate that the emplacement age of the Duoni Formation volcanic flows is 120.2 ± 0.5 Ma. Paleomagnetic results from 235 progressively demagnetized volcanic rock samples (25 sites) and 41 sandstone samples (5 sites) indicate that the dominant remanence carriers are Ti-poor titanomagnetite and Ti-poor titanohematite in the volcanic samples and Ti-rich titanomagnetite in the sandstone samples. Rock magnetic investigations, systematic demagnetization behavior, positive fold test results, and direct petrographic identifications all indicate that the paleodirections recorded by the chemically stable magnetic particles are primary thermal remanent magnetization in the volcanic flows and primary detrital remanent magnetization in the sandstones. The tilt-corrected ChRM mean direction is $D/I = 356.4°/16.4°$ with $\alpha_{95} = 6.3°$ ($N = 19$), corresponding to a paleopole position of $\lambda_p = 66.9°N$, $\varphi_p = 281.2°E$ with $A_{95} = 6.1°$. Combined with previously published results, the geochronological dating and paleomagnetic analysis indicate a paleolatitude of 13.1 ± 2.7°N for the southern margin of the Lhasa block during the Early Cretaceous. Therefore, the southernmost margin of the Eurasian continent likely remained at the low-middle paleolatitude of 13.1 ± 2.7°N between the Early Cretaceous and the Paleocene. Based on comparisons to results from the Tethyan–Himalayan block and the reference poles from stable India and Eurasia, the low-middle paleoposition of the Lhasa block during the Early Cretaceous through Early Paleocene suggests that the initial contact between India and Asia occurred at ca. 59.3 Ma. Under the assumption of a rigid Eurasian plate, this timing implies that a total collision-related latitudinal convergence of 1450 ± 400 km (13.1 ± 3.7°N) has been accommodated by folding, thrust faulting, normal faulting, crustal thickening, intracontinental subduction in Tibet and central Asia and southeastward continental extrusion of the Indo-China block from the eastern syntaxis between the Lhasa block and stable Asia.

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1. Introduction

The collision between the Indian and Eurasian continents during the Early Cenozoic was largely responsible for the formation of the Himalayan orogen and the uplift of the Tibetan Plateau (Chang and Zhen, 1973; Allégre et al., 1984; Dewey et al., 1988, 1989; Beck et al., 1995; Yin and Harrison, 2000; Ding et al., 2005; Cai et al., 2011; Zhang et al., 2012; Ding et al., 2014; Zhang et al., 2014; Xu et al., 2015; Zhang et al., 2017). Although the tectonic settings and tectonostratigraphic divisions of the Himalaya and the Tibetan Plateau have the subject of intense research in the Earth science community in recent years, multiple scientific questions still remain matters of debate. For instance, the precise timing of the initiation of India–Asia collision is highly controversial due to different estimates based on different types of evidence, and these estimates range from 70 Ma to 34 Ma (*e.g.*, Jaeger et al., 1989; Rowley, 1998; Yin and Harrison, 2000; Ding et al., 2005; Aitchison et al., 2007; Chen et al., 2010; Najman et al., 2010; Sun et al., 2010; Tan et al., 2010; Aitchison et al., 2011; Meng et al., 2012; Sun et al., 2012; van Hinsbergen et al., 2012; Gibbons et al., 2015; Hu et al., 2016; Jiang et al., 2016; Wang et al., 2017). The newest paleomagnetic work conducted on Eocene sediments deposited in the Xigaze forearc basin in southern Tibet support a ~50 Ma Greater India–Asia collision at ~24°N (Meng et al., 2012). Based on the latest compilation of paleomagnetic data from the southern margin of Eurasia, India and the Himalaya, a new two-stage collision model has been proposed and is still hotly debated (van Hinsbergen et al., 2012). This subject is clearly a significant starting point for discussing the formation of the Himalayan–Tibetan orogenic system and associated plateau uplift.

Plateau uplift is closely related to Tibetan over-thickened crust associated with the India–Asia collision. Scholars have developed several

* Corresponding author. Tel.: +86 10 8409 7104.
E-mail address: lizy@itpcas.ac.cn (Z. Li).

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end-member mechanisms to account for this crustal thickening, *e.g.*, continental shortening triggered by folding and thrusting distributed across Tibet (Dewey and Burke, 1973; Chang et al., 1986; Dewey et al., 1988), under-thrusting of Indian crust beneath Asia (Zhao et al., 1993; Hodges, 2000), and lateral extrusion of continental crust (Tapponnier and Molnar, 1979; Tapponnier et al., 1982). The thickening of the Tibetan crust depends on the total amount of convergence partitioned between India and Asia. Thus, accurate estimates of the total amount of convergence are critical for constraining these models. Over the past two decades of active research, researchers using different types of evidence have proposed a wide range of values for the total amount of post-collisional crustal shortening between northern India and Asia to reconcile geological observations with geophysical and paleomagnetic data (Achache et al., 1984; Patriat and Achache, 1984; DeCelles et al., 2002; Johnson, 2002; Sun et al., 2008; Chen et al., 2010; Dupont-Nivet et al., 2010a,b; Liebke et al., 2010; Sun et al., 2010; Chen et al., 2012; Meng et al., 2012; Sun et al., 2012; van Hinsbergen et al., 2012). Among the available paleomagnetic data sets, the estimates of the paleolatitude of the Lhasa block vary from 5°N to 30°N for the period between the Early Cretaceous and the Early Paleocene (Achache et al., 1984; Lin and Watts, 1988; Chen et al., 1993a,b; Liebke et al., 2010; Tan et al., 2010). Based on different lines of evidence, investigators have proposed different crustal shortening estimates for Tibet and stable Asia since the onset of the India–Asia collision (Achache et al., 1984; Patriat and Achache, 1984; DeCelles et al., 2002; Johnson, 2002; Sun et al., 2008; Chen et al., 2010; Dupont-Nivet et al., 2010a,b; Liebke et al., 2010; Sun et al., 2010; Chen et al., 2012; Meng et al., 2012; Sun et al., 2012; van Hinsbergen et al., 2012). For instance, paleogeographic reconstructions based on paleomagnetic data from the Himalayan orogen and the Lhasa block suggest that 1500–3000 km or more of post-collisional convergence has taken place between Greater India and Eurasia, with 1100–2000 km of this total amount accommodated by north-south shortening in the Tibetan Plateau and stable Asia (Achache et al., 1984; Patriat and Achache, 1984; Chen et al., 2010; Dupont-Nivet et al., 2010a,b; Liebke et al., 2010; Sun et al., 2010, 2012; van Hinsbergen et al., 2012). However, recent paleomagnetic studies can be divided into two groups on this subject: One group of scholars argues for relatively low amounts of crustal shortening, *e.g.*, 360 ± 280 km to 800 ± 600 km based on Upper Cretaceous sediments and lava flows (Tan et al., 2010; Chen et al., 2012; Meng et al., 2012) and up to 870 ± 520 km based on Lower Cretaceous lava flows (Tan et al., 2010; Chen et al., 2012; Meng et al., 2012). The other group of scholars favors much higher amounts of continental shortening, with estimates ranging from 1100 km to over 2000 km (Chen et al., 2010; Dupont-Nivet et al., 2010a,b; Liebke et al., 2010; van Hinsbergen et al., 2012; Tang et al., 2013). Based on several lines of geologic evidence, *e.g.*, balanced cross-sections, volumetric analyses and the rates of convergence between the two continents (Johnson, 2002), certain geologists agree with the higher estimates based on paleomagnetic research (Dewey et al., 1988, 1989; Ratschbacher et al., 1994; Yin and Harrison, 2000; Johnson, 2002).

Clearly, accurately identifying the paleoposition of the Lhasa block is important to precisely determining the timing of initial contact between India and Asia and the total amount of convergence between the two continents. As a powerful quantitative technique, paleomagnetism has played an important role in the study of the paleolatitude of the Lhasa block prior to the collision between India and the southern margin of the Lhasa block. However, additional Cretaceous results are needed to further tackle this scientific question.

Therefore, this study focuses on geochronological and paleomagnetic research. A study area in the northern part of the Lhasa block (central Tibet) was selected to avoid both the intense magmatic activity along the southern margin of the Lhasa block during the Early Cenozoic and the strong tectonic deformation present in Cretaceous and lower Cenozoic outcrops.

In this paper, we report both zircon U–Pb dating and paleomagnetic results from the Duoni Formation volcano-sedimentary sequences near Luoma town in Nagqu County, central Tibet. We attempt to determine the following: (1) the precise emplacement ages of the volcanic rocks in the Duoni Formation; (2) the paleoposition of the Lhasa block prior to the collision of India and Asia and, in particular, the paleolatitude of the Lhasa block during the Early Cretaceous; and (3) the total amount of post-collisional latitudinal shortening that has occurred across Tibet and stable Asia based on the comparison of our results and previous work.

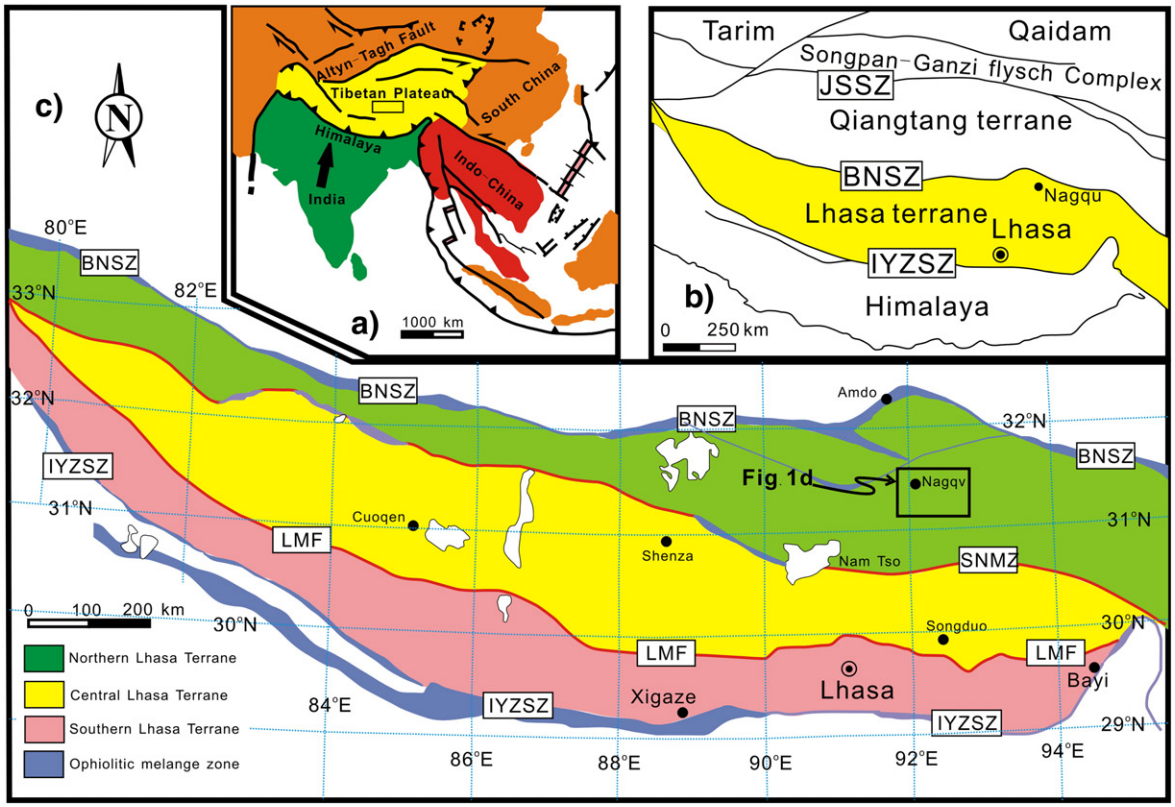
2. Geological setting and sampling

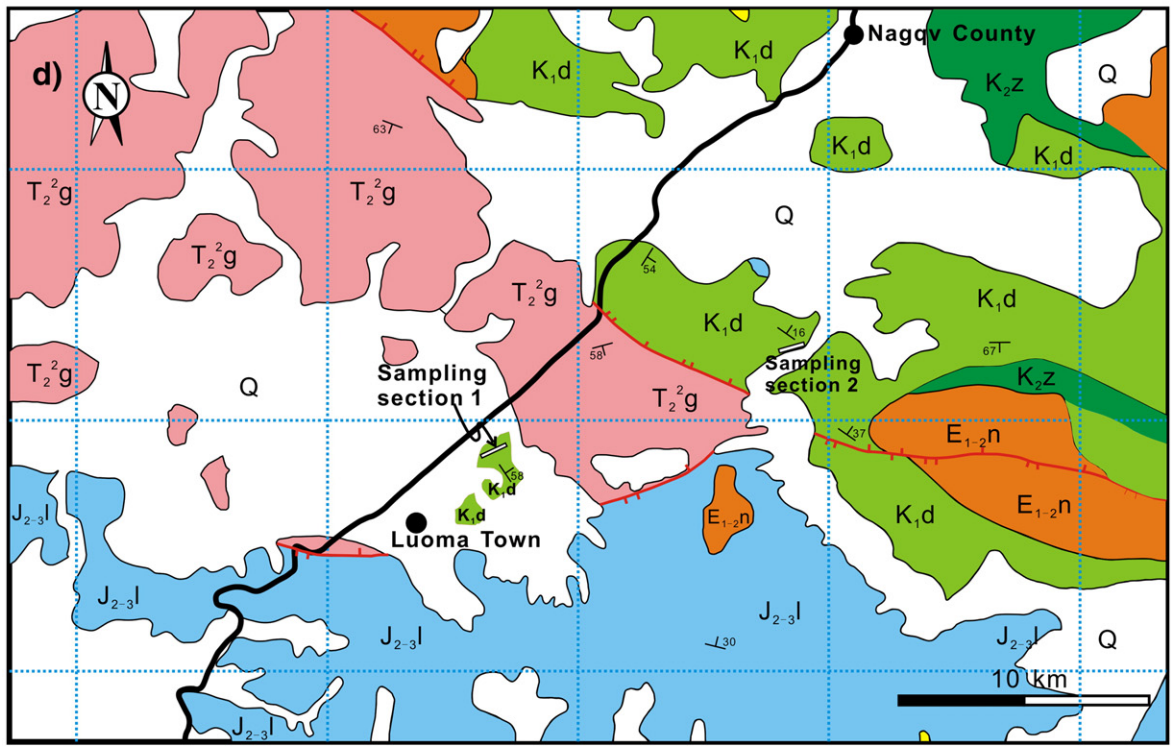
The Himalayan–Tibetan orogenic belt is composed of a series of E–W-trending blocks that progressively accreted onto Asia during the late Paleozoic and Mesozoic eras, prior to the India–Asia collision (Chang and Zhen, 1973; Dewey et al., 1988; Yin and Harrison, 2000) (Fig. 1a, b). These blocks are, from north to south, the Songpan–Ganze–Hoh-Xil block, the Qiangtang block, the Lhasa block, and the Tethyan–Himalayan block (Yin and Harrison, 2000). These blocks are separated by the Jinsha suture zone (JSSZ), the Bangong–Nujiang suture zone (BNSZ), and the Indus-Yarlung Zangbo suture zone (IYZSZ), which represent the closure of the Paleo-Tethyan, Meso-Tethyan and Neo-Tethyan oceans, respectively (Yin and Harrison, 2000) (Fig. 1b). The accretion of the Lhasa block (Fig. 1b,c) predates the arrival of India and represented the southernmost margin of Asia prior to the India–Asia collision (Sengör, 1987; Dewey et al., 1988; Yin and Harrison, 2000; Kapp et al., 2005, 2007; Leier et al., 2007; Zhang et al., 2011, 2012, 2014). The block is bounded by the BNSZ to the north and the IYZSZ to the south (Chang and Zhen, 1973; Allégre et al., 1984; Dewey et al., 1988; Yin and Harrison, 2000; Zhang et al., 2012, 2014; Chen et al., 2017) (Fig. 1b,c). The N–S width of the Lhasa block is approximately 300 km at the longitude 91°E, near Lhasa, and narrows to less than 100 km at the longitude of 80°E, near the Shiquanhe area in western Tibet (Fig. 1c). The southern portion of the Lhasa block consists primarily of Paleozoic and Mesozoic marine strata and arc-type volcanic rocks that are intruded by Jurassic through lower Tertiary granitoids of the Gangdese batholith (Yin and Harrison, 2000). The northern portion consists primarily of Jurassic–Cretaceous sedimentary and igneous rocks (Leeder, et al., 1988; Zhang, 2004; Pan and Ding, 2005; Leier et al., 2007; Zhang et al., 2007; Zhu et al., 2008) that form an E–W-trending belt of volcano-sedimentary sequences (Ma and Yue, 2010).

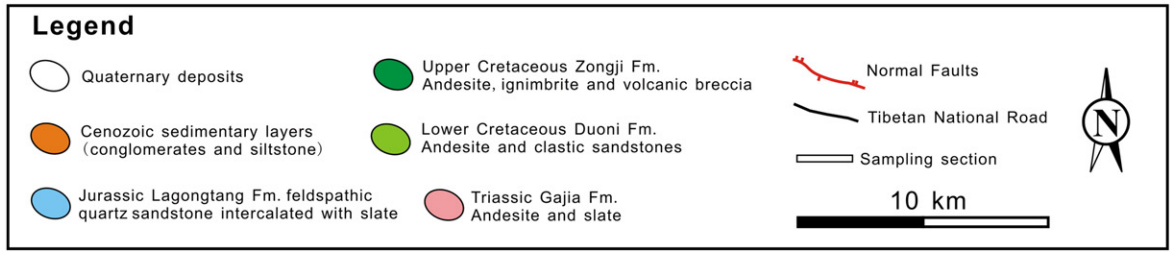
The study area, bounded by the Bangong–Nujiang suture to the north and the Nyianqing–Tanggula graben to the south, is located in the northern part of the Lhasa block (Fig. 1a–d). Several suites of rock units are present in this area (*e.g.*, the Middle-Upper Jurassic Lagongtang Formation, the Lower Cretaceous Duoni Formation, and the Upper Cretaceous Zonggei Formation) (Fig. 1d).

The Lagongtang Formation (J_{2-3}) is overlain disconformably by the volcano-sedimentary sequences of the Duoni Formation (K_1d), which is dominated by slate, chert, and medium- to fine-grained detrital quartz sandstone, and most of these layers are folded (Fig. 2). In certain places, andesite and basaltic andesite units crop out (volcanic lava flows are intercalated with 3–4 sandstone layers in section 1 in this study) (Figs. 1d and 2a–f). Several different types of fossils, such as *Pernidella* sp., *Spongiom orphan* sp., and *Coniopteris* sp., have been found in the sedimentary layers of the Duoni Formation (BGMRX, 1993; Regional Geological Survey Report, 2005). Furthermore, this formation also contains abundant plant fossils, such as ferns and cycads, including *Weichselia reticulata*, a global standard type fossil with a characteristic Lower Cretaceous age. Therefore, according to strata correlation with adjacent basins and fossil identification, the Duoni Formation is Lower Cretaceous in age.

Higher in the section, the Duoni Formation is unconformably overlain by the Zonggei Formation (K_2z), which is dominated by andesite and andesitic lithic volcanic breccia. This formation is commonly thought to have formed in a convergent setting closely related to the northward subduction of the Neo-Tethys oceanic seafloor during the Late Cretaceous (BGMRX, 1993; Regional Geological Survey Report, 2005).


c)
N
Altyn Tagh Fault
Tibetan Plateau
South China
Himalaya
India
Indo-China
a)
1000 km
Tarim
Qaidam
Songpan-Ganzi flysch Complex
JSSZ
Qiangtang terrane
BNSZ
Nagqu
Lhasa terrane
Lhasa
IYZSZ
b)
Himalaya
0 250 km
80°E
82°E
33°N
32°N
31°N
30°N
29°N
84°E
86°E
88°E
90°E
92°E
94°E
Amdo
Fig 1d
Nagqv
Cuoqen
Shenza
Nam Tso
SNMZ
Songduo
Bayi
Xigaze
LMF
IYZSZ
0 100 200 km
Northern Lhasa Terrane
Central Lhasa Terrane
Southern Lhasa Terrane
Ophiolitic melange zone


d)
N
Nagqv County
Luoma Town
Sampling section 1
Sampling section 2
K1d
K2z
Q
T2²g
E1-2n
J2-3l
10 km


Legend
Quaternary deposits
Cenozoic sedimentary layers (conglomerates and siltstone)
Jurassic Lagongtang Fm. feldspathic quartz sandstone intercalated with slate
Upper Cretaceous Zongji Fm. Andesite, ignimbrite and volcanic breccia
Lower Cretaceous Duoni Fm. Andesite and clastic sandstones
Triassic Gajia Fm. Andesite and slate
Normal Faults
Tibetan National Road
Sampling section
10 km
N

In the field, the dip angles of the folded strata range from 30° to 70°. The volcanic lava flows and sedimentary sandstone layers of the Duoni Formation (investigated during the sampling campaign) dip to the northeast in the study area (*i.e.*, trending E–W and NW–SE) (Fig. 2c, g; Table 1). In section 1 (Fig. 1d), volcanic lava flows are freshly exposed throughout the section and are dominated by andesite and basaltic andesite, with at least 3–4 intercalated sandstone layers, each of which is 1–2 m thick. These layers dip to the northeast with dip angles of 50–70°, which is regionally consistent with the bedding of the folded strata. The bedding attitudes of the volcanics are easy to determine due to presence of the sandstone interlayers. In section 2, approximately 8 km to the northeast of section 1 (Fig. 1d), the sedimentary layers are dominated by sandstone and siltstone layers that dip to the northeast (55–58°) with shallower dip angles (<20°) than those in section 1 (Fig. 1d; Table 1).

Based on strata correlation, fossil identification, and structural analysis, the folding age of the Duoni Formation can be constrained to before the Late Cretaceous but after/during the Early Cretaceous (*i.e.*, ~120 Ma). This age will help us place a tight constraint on the remanence acquisition age of the paleodirections recorded in the strata.

Several scholars have dated the volcanics of the Duoni Formation via the zircon U–Pb dating method and obtained Lower Cretaceous ages (XIGS, 2005; Liu et al., 2011). We selected well-exposed volcanic outcrops and performed paleomagnetic and geochronological sampling to test the validity of the dating results in the study area. Field expeditions near Nagqu County were separately conducted in 2011 and 2012.

In the 2011 field season, 2 bulk samples for geochronological dating and a total of ~250 paleomagnetic samples (labeled TL) were collected from freshly exposed volcanics (andesite and basaltic andesite) at 25 evenly distributed paleomagnetic sites in section 1 (Fig. 2a–f; Table 1). In the 2012 field campaign, a total of 43 sandstone and siltstone samples (labeled TND) were collected from 5 paleomagnetic sites in section 2 (Fig. 2g,h; Table 1). The two sampling campaigns were conducted near Luoma town, Nagqu County, approximately 100 km north of the city of Lhasa (Fig. 1c,d).

At each site, 8 to 10 independently oriented paleomagnetic samples were drilled in an area approximately 1–2 m wide using a portable gasoline-powered drill. All samples were oriented *in situ* using both magnetic and sun compasses (if local weather allowed). The differences between the readings of the two compasses were limited to ±2°; hence, corrections were not necessary. The present geomagnetic field (PGF) direction and present dipole field (PDF) direction were calculated to be Dec = 0.0°, Inc = 48.3°, and Dec = 0.0°, Inc = 50.6°, respectively, at the sampling site (31.3°N, 91.9°E).

3. Zircon U–Pb age dating

3.1. Sample selection and analytical methods

Two bulk samples (2011TS27-32 and 2011TS27-14) of andesite for geochronological dating were sampled from the volcanic flows located near Luoma town for this study. Due to the limited number of zircon grains obtained from sample 2011TS27-14, the results from this sample are treated as a reference.

The two samples were processed via heavy liquid and magnetic separation methods at the Center for Rocks and Minerals Separation in Langfang city. The separated zircon grains from each sample were manually picked, mounted on adhesive tape, enclosed in epoxy resin, polished to approximately half of their initial width and photographed under reflected and transmitted light. Cathodoluminescence (CL) images of the zircon grains from each established target were taken using a Mini CL detector (Gatan Co. USA) attached to an Electron Microprobe (JXA-8100, JEOL, Japan) at the Institute of Geology and Geophysics, Chinese Academy of Sciences (IGGCAS), Beijing. The mount was then cleaned and LA-ICP-MS analyses were performed on the polished mount surfaces using an Nd-YAG 213 laser ablation system (Microprobe 2, New Wave Research, USA) coupled with a VG PQ Excel ICP-MS. The detailed analytical procedures and parameters are described in Ding et al. (2014). In our analysis, the zircon standards PLE and SL were used as calibration standard samples. The laser featured a repetition rate of 10 Hz, an energy of 8–10 J/cm^2, and a focused beam diameter of 20–35 μm, depending on the grain size of the samples' zircons.

The isotopes ^{204}Pb, ^{206}Pb, ^{207}Pb, ^{232}Th and ^{238}U were measured, and the ages were calculated from the U and Th decay constants reported by Steiger and Jäger (1977). All reported ages were calculated using Isoplot 3.00 (Ludwig, 2003). Individual analyses in the data table and the concordia plot are presented with 1σ error, and the uncertainties in ages are presented at the 95% confidence level (2σ). Our results also show that all the data plot on or near the concordia curve; thus, no Pb loss is present in our samples (Fig. 3; Supplementary Table A1).

3.2. Zircon U–Pb analytical results

The samples 2011TS27-32 and 2011TS27-14 were collected from the volcanic ramp. Sample 2011TS27-32 was collected from the lower part of the emplacement ramp in section 1, and sample 2011TS27-14 was collected from the upper part of the emplacement ramp in section 1. The majority of the zircon grains separated from these two samples are euhedral to subhedral. The selected zircons are ~100 μm long and are nearly equigranular and slab shaped. Dim internal structure in the zircons is evident in the CL images. A limited number of grains are euhedral with comparatively clear oscillatory zoning, which indicates a magmatic origin.

As shown in Fig. 3, the 24 zircon grains analyzed from sample 2011TS27-32 yield a concordant age of 120.2 ± 0.5 Ma (MSWD = 0.93) (Fig. 3a,b; Supplementary Table A1).

In contrast to sample 2011TS27-32, sample 2011TS27-14 yielded 7 zircons grains, which produce a concordant age of 123.4 ± 2.4 Ma (MSWD = 0.99) (Fig. 3c,d; Supplementary Table A1). This result is consistent with the dating result from 2011TS27-32 within error, even though fewer zircon grains were analyzed (Fig. 3; Supplementary Table A1).

In summary, the combined results of samples 11TS27-32B and 11TS27-14 tightly constrain the formation of the volcanic lava flows in the Duoni Formation to 120 Ma.

Fig. 1. (a) Schematic tectonic framework of the Tibetan Plateau and adjacent areas. The study area, indicated by the rectangle, is located in the northern part of the Lhasa block in central Tibet. (b) Tectonic outline of the Tibetan Plateau (after Zhu et al., 2008). The yellow-colored area illustrates the extent of the Lhasa block. (c) Tectonic framework of the Lhasa block, showing the major tectonic subdivisions (Zhu et al., 2009). The rectangle indicates the position of the study area shown in Fig. 1d. (d) Geologic map of the study area near Luoma town, Nagqu County (scale: 1:250,000). Abbreviations are as follows: JSSZ = Jinsha suture zone; BNSZ = Bangong–Nujiang suture zone; IYZSZ = Indus–Yarlung Zangbo suture zone; SNMZ = Shiqunhe–Nam Tso Mélange Zone; LMF = Luobadui–Milashan Fault. The green-, yellow-, and pink-colored areas are the Northern Lhasa subblock, Central Lhasa subblock and the Southern Lhasa subblock, respectively, and the linear light blue zones represent ophiolitic mélange zones.

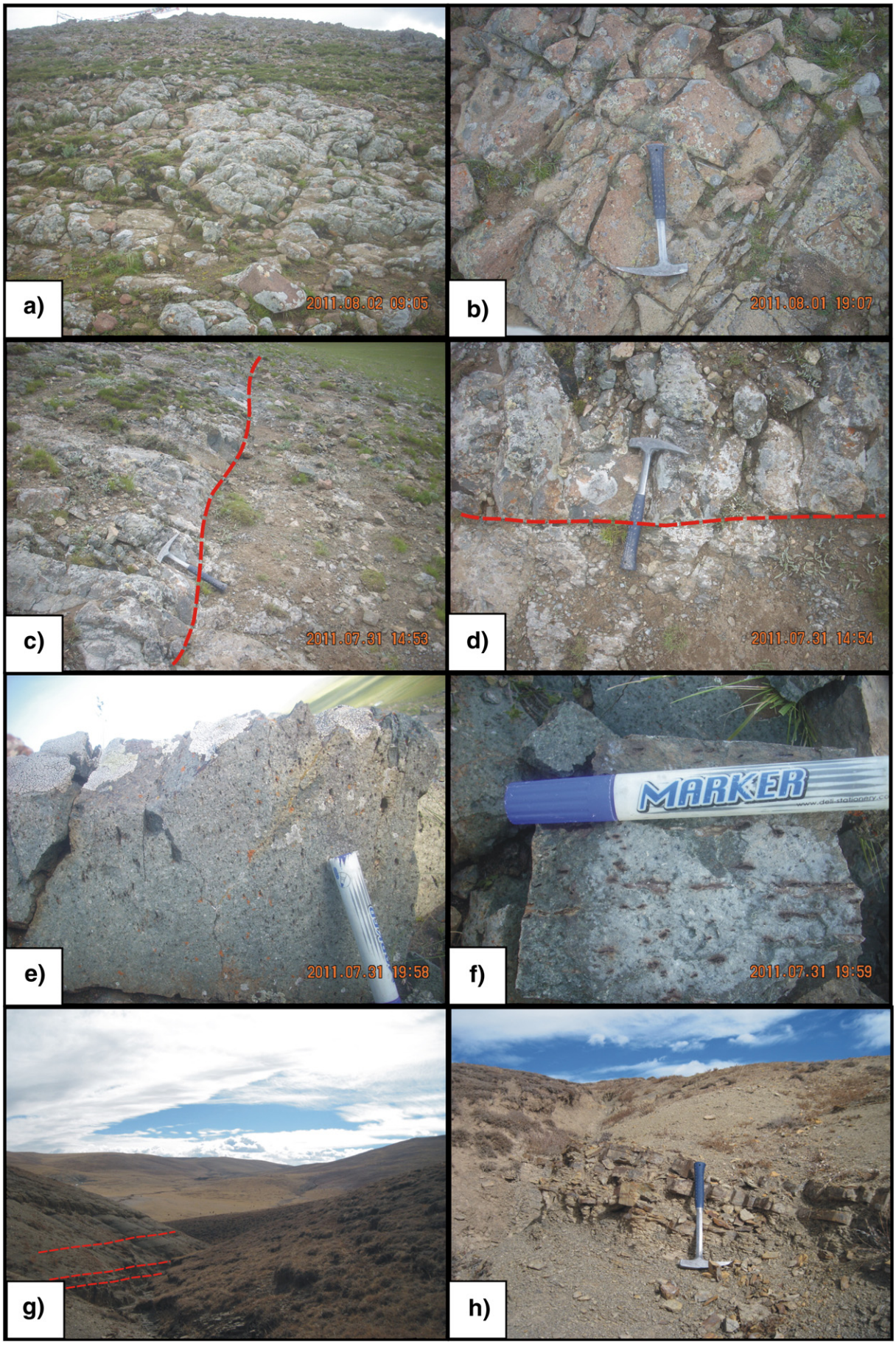

a)
2011.08.02 09:05
b)
2011.08.01 19:07
c)
2011.07.31 14:53
d)
2011.07.31 14:54
e)
2011.07.31 19:58
f)
MARKER
2011.07.31 19:59
g)
h)

Table 1
Summary of all ChRM site-mean directions from volcanic and sandstone specimens from the Duoni Fm. near Nagqu county.

Site ID	Lithology	Strike/dip (°)	n/no (N)	R/N	Dg (°)	Ig (°)	Ds (°)	Is (°)	kg/ks	α95g/α95s (°)
Volcanic samples										
TL01	Basaltic andesite	327/57	4 (1)/10	0/3	335.3	25.5	352.7	7.2	721.1/714.9	3.6/3.6
TL02[a]	Basaltic andesite	320/58	11/11	11/0	355.5	18.5	356	−17.4	147.1/147.9	3.8/3.8
TL03[a]	Basaltic andesite	325/57	10/10	8/2	356.9	27.3	5.3	−8.3	58.4/58.6	6.4/6.4
TL04	Andesite	326/58	9/9	0/9	325	28.1	349.9	15.2	62.5/62.4	6.6/6.6
TL05	Andesite	332/60	7/8	5/2	158.7	−41.9	192.1	−15	82.5/82.7	6.7/6.7
TL06	Andesite	315/57	8/9	0/8	303.1	42.1	348.4	29.6	24.9/24.9	11.3/11.3
TL07	Andesite	315/57	5/8	5/0	123.7	−21.9	148.3	−20.8	194.4/191.4	5.5/5.5
TL08	Andesite	315/57	8 (6)/10	0/2	322.5	36.4	349.8	13.6	53.7/53.5	8.3/8.3
TL09[b]	Andesite	315/57	5/10	5/0	14.1	−5.3	352.2	−50.1	190.9/191.5	5.6/5.5
TL10	Andesite	315/57	9/9	1/8	317.8	28.6	340.8	13	31.5/31.5	9.3/9.3
TL11	Basaltic andesite	315/57	5/10	0/5	315.5	17.6	330.2	9.1	65.8/65.8	9.5/9.5
TL12	Basaltic andesite	315/57	10 (5)/10	0/5	346.5	54.2	14.5	10.7	44.5/44.6	7.6/7.5
TL13[c]	Basaltic andesite	317/57	(7)/10	0/0	/	/	/	/	/	/
TL14[c]	Andesite	317/57	(1)/9	0/0	/	/	/	/	/	/
TL15	Andesite	317/57	9 (6)/9	0/3	355	55.5	20.1	9	80.8/80.8	6.1/6.1
TL16[c]	Andesite	335/64	0/9	0/0	/	/	/	/	/	/
TL17[a]	Andesite	335/64	5/10	5/0	202.1	−21	199.4	27.3	75.9/76.6	8.8/8.8
TL18	Andesite	335/64	9/9	0/9	329.5	46.7	17.4	22.2	50.3/50.2	7.3/7.3
TL19[c]	Andesite	335/64	2/9	1/1	/	/	/	/	/	/
TL20[c]	Andesite	335/64	1/9	0/1	/	/	/	/	/	/
TL21[c]	Basaltic andesite	335/64	2/8	0/2	/	/	/	/	/	/
TL22	Basaltic andesite	335/64	3/10	0/3	340	32	6	9.5	105.9/106.0	12.0/12.0
TL23[a]	Andesite	335/64	7/10	7/0	8.7	33.9	20.5	−9.7	29.9/29.9	11.2/11.2
TL24	Basaltic andesite	335/64	8/10	0/8	332.7	24	356	12.2	78.7/78.7	6.3/6.3
TL25	Basaltic andesite	335/64	6 (1)/9	2/3	326.4	31.4	1.1	20.1	35.6/35.5	11.6/11.6
	Volcanic mean		14	2/12	326.8	35.5	356.3	15.4	24.0/22.5	8.3/8.6
Paleopole Lat. = 66.3°, Long. = 281.1°, K = 23.6, A95 = 8.4°, N = 14										
Sandstone samples										
TND01	Siltstone	325/16	7/8	0/7	342.6	22.9	348.1	17.3	42.1/42.2	9.4/9.4
TND02	Siltstone	325/16	8/8	0/8	353	35.7	1.3	27.1	52.8/52.8	7.7/7.7
TND03	Sandstone	328/16	8/8	0/8	352.3	26.8	358.4	19.4	42.9/42.9	8.6/8.6
TND04	Sandstone	328/16	7/8	0/7	354.7	24.3	359.9	16.4	69.7/69.6	7.3/7.3
TND05	Sandstone	328/16	9/9	0/9	349.7	22.2	354.7	15.6	68.6/68.7	6.3/6.3
	Sandstone mean		5	0/5	350.4	26.4	356.4	19.2	135.3/141.0	6.6/6.5
Paleopole Lat. = 68.4°, Long. = 281.4°, K = 194.3, A95 = 5.5°, N = 5										
	Combined mean		19	2/17	333.5	33.5	356.4	16.4	21.4/29.3	7.4/6.3
Paleopole Lat. = 66.9°, Long. = 281.2°, K = 31.4, A95 = 6.1°, N = 19										
Fold test results										
(1) McFadden's correlation test (1990): N = 19, Xi2 = 9.397 (=3.820) before (after) bedding correction. The critical value, Xi_c, is 5.075 and 7.112 at the 95% and 99% confidence levels respectively.										
(2) Watson and Enkin's unfolding test (1993): The maximum value of precision parameter K is achieved at the 79.1 ± 14.1% unfolding level.										

Abbreviations are as follows: Site ID, Site Identification; Strike/dip, Right-hand strike azimuth and dip angle of the bed; n/n_o, number of samples used in calculating Fisher mean/number of demagnetized samples; R/N, number of ChRM directions that display normal/reversed polarity in calculating Fisher site-means; Dg, Ig (Ds, Is), declination and inclination of an isolated ChRM direction before (after) bedding correction; kg (ks), estimates of precision parameter of Fisher mean before (after) correction; α95g (α95s), radius of 95% confidence circle about the calculated mean direction of an *in situ* (tilt-corrected) coordinate; Lat., Long., latitude and longitude of paleopole obtained in different groups; A95, radius of the 95% confidence circle about the calculated mean pole; K, estimate of the precision parameter about the mean paleopole obtained in this study.

[a] Site was discarded from further analysis due to probable excursion recordings of the geomagnetic field.

[b] Site was excluded from further analysis due to probable disturbance.

[c] Site was discarded before calculating final mean direction because the ChRM directions within site were outliers in the iterative process of data selection. See details in the text.

4. Mineralogical investigation

4.1. Thermomagnetic analysis

A residual slice sample from each site was selected for rock magnetic measurements to identify the dominant ferro-/ferrimagnets and monitor possible chemical changes and mineralogical phase transformations during a heating–cooling cycle in the thermomagnetic analysis (Fig. 4a, b). We measured the temperature dependence of the susceptibility curve for representative samples, and all heating–cooling cycles were conducted with an MFK-1 device coupled to a CS-4 temperature control apparatus in an air atmosphere (the interaction of oxygen and heating samples may result in chemical changes). All samples exhibit a noticeable decrease in magnetic susceptibility near the Curie point of stoichiometric magnetite (~585 °C) during the heating–cooling cycles (Fig. 4a, b). Additionally, all samples, except sample TL2-5, feature heating curves that are slightly but not significantly higher than the cooling curves, which could be attributed to a small quantity of strong magnetic minerals, such as

Fig. 2. Photographs taken in the field near Nagqu County in this study. (a) Photo showing volcanic outcrops in section 1 (Fig. 1d) near Luoma town. View is to the northeast. (b) Freshly exposed rocks in an andesite outcrop. The rock is suitable for paleomagnetic research and geochronologic dating. Hammer for scale. (c) Contact between overlying volcanic rocks (on the left) and intercalated sandstone layers (on the right). The hammer straddles the contact boundary, and the red line delineates the boundary of the contrasting lithologies. View is to the east. (d) Enlargement of photo (c) and close examination of the boundary between the volcanic rocks (upper part) and sandstone rocks (lower part). Hammer for scale. (e) The pore-like texture of the volcanic rocks is filled with calcite. Marker (approximately ~15 cm) for scale. (f) Close examination of the pore-like texture found inside the volcanic rocks. Marker (approximately ~15 cm) for scale. (g) Sandstone layers of the Duoni formation. The layers unconformably rest on volcanic flows, and the red lines indicate the right-hand strike direction of sandstone layers. View is to the southeast. (h) Close examination of the exposed sandstone layers, which dip to the northeast with low-medium dip angles. View is to the north, and hammer for scale.

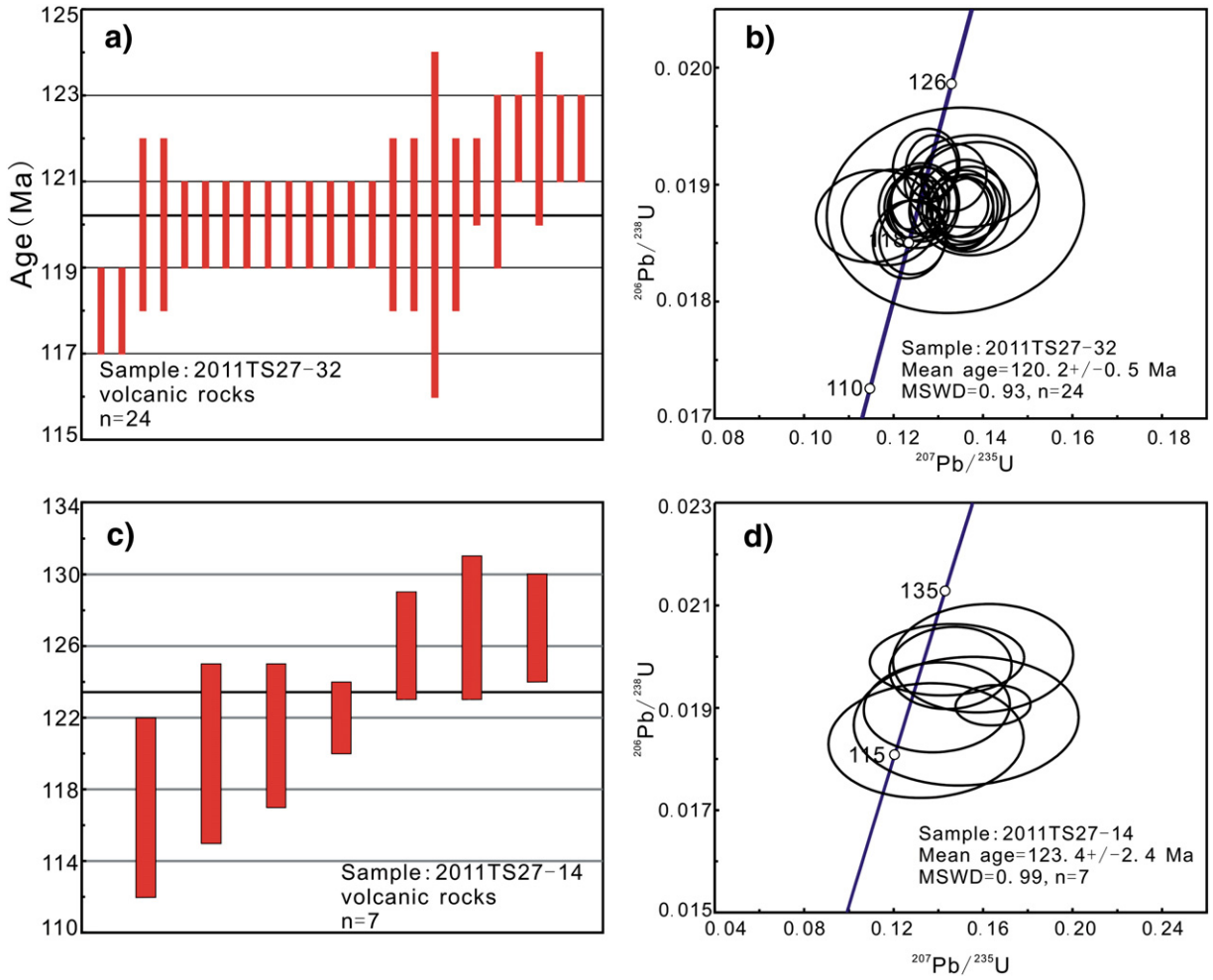


Fig. 3. Weighted mean age (left) and concordia (right) plots of zircon grains from the geochronological samples in this study. (a) Weighted mean age plot for U–Pb dating results from sample 11TS27-32, collected from the upper part of the volcanic emplacement in section 1 (see Fig. 1d). (b) Concordia plot of U–Pb dating results from 24 zircons. (c) Weighted mean age plot for U–Pb dating results from sample 11TS27-14, collected from the lower part of the volcanic emplacement in section 1 (see Fig. 1d). (d) Concordia plot of U–Pb dating results from 7 zircons. The dating results of 11TS27-14 were only used as a reference in this study. See Supplementary Table A1 for details on the U–Pb isotopic compositions and U–Pb dating results.

magnetite, being produced during the heating–cooling cycle (Dunlop and Özdemir, 1997). All selected samples exhibit heating curves with a susceptibility peak near ~300 °C (*e.g.*, Fig. 4a, b). The steady increase in susceptibility below 300 °C may be ascribed to gradual unblocking of fine-grained (near the superparamagnetic/single-domain (SP/SD) boundary) ferromagnetic particles. This susceptibility increase may also result from the newly produced magnetic minerals with strong saturated magnetization and comparatively low coercivity (*e.g.*, the conversion of metastable maghemite to stable magnetite), which has been observed in secondary maghemite grains formed by pedogenesis in Chinese loess (Deng et al., 2006), or may result from the exsolution of titanomagnetite into magnetite with comparatively higher susceptibility than Ti-rich magnetite during the heating process (Dunlop and Özdemir, 1997).

Here, we postulate that the exsolution of Ti-poor titanomagnetite into magnetite (with high magnetization) and titanite is the probable cause for the steady increase in susceptibility at temperatures lower than 300 °C (Dunlop and Özdemir, 1997).

4.2. Microscopic observations (polarized- and reflected-light observations)

Microscopy (using polarized and reflected light) allowed us to directly identify mineral types (*e.g.*, plagioclase, pyroxene, and magnetite particles), particle shapes, sizes, and the minerals' distribution throughout the bulk samples.

Typical samples were selected for polarized- and reflected-light microscope observations at the Key Laboratory of Continental Collision and Plateau Uplift (LCPU), Institute of Tibetan Plateau Research, Chinese Academy of Sciences (ITPCAS), Beijing. In the photographs shown in Fig. 4c and d, certain common crystalline rock-forming minerals, such as feldspar (bright white spots) and hornblende (bright green- or blue-colored spots), are easily identified. Magnetic minerals (*e.g.*, magnetite) can be identified as bright particles under reflected light due to their opaque properties and high reflectivity characteristics (Fig. 4c, d).

In our samples, magnetite particles clearly coexist with feldspar and hornblende (Fig. 4c, d). We do not observe any corrupted mineral margins in thin section, which indicates that the volcanic lava flows have not experienced even low-grade metamorphism and/or severe weathering. Based on the thermomagnetic analysis and observations, we conclude that the magnetic minerals very likely crystallized during the cooling of the magma and are therefore primary. Thus, Ti-poor titanomagnetite and Ti-poor titanohematite particles are most likely the primary remanence carriers for the volcanic lava flows in the Duoni Formation.

Based on the results in "Paleomagnetic methods and results" (Section 5 below) and the mineralogical investigations discussed above, the characteristic remanent magnetization (ChRM) remanences of both the volcanic rocks and sandstone layers are of primary origin.

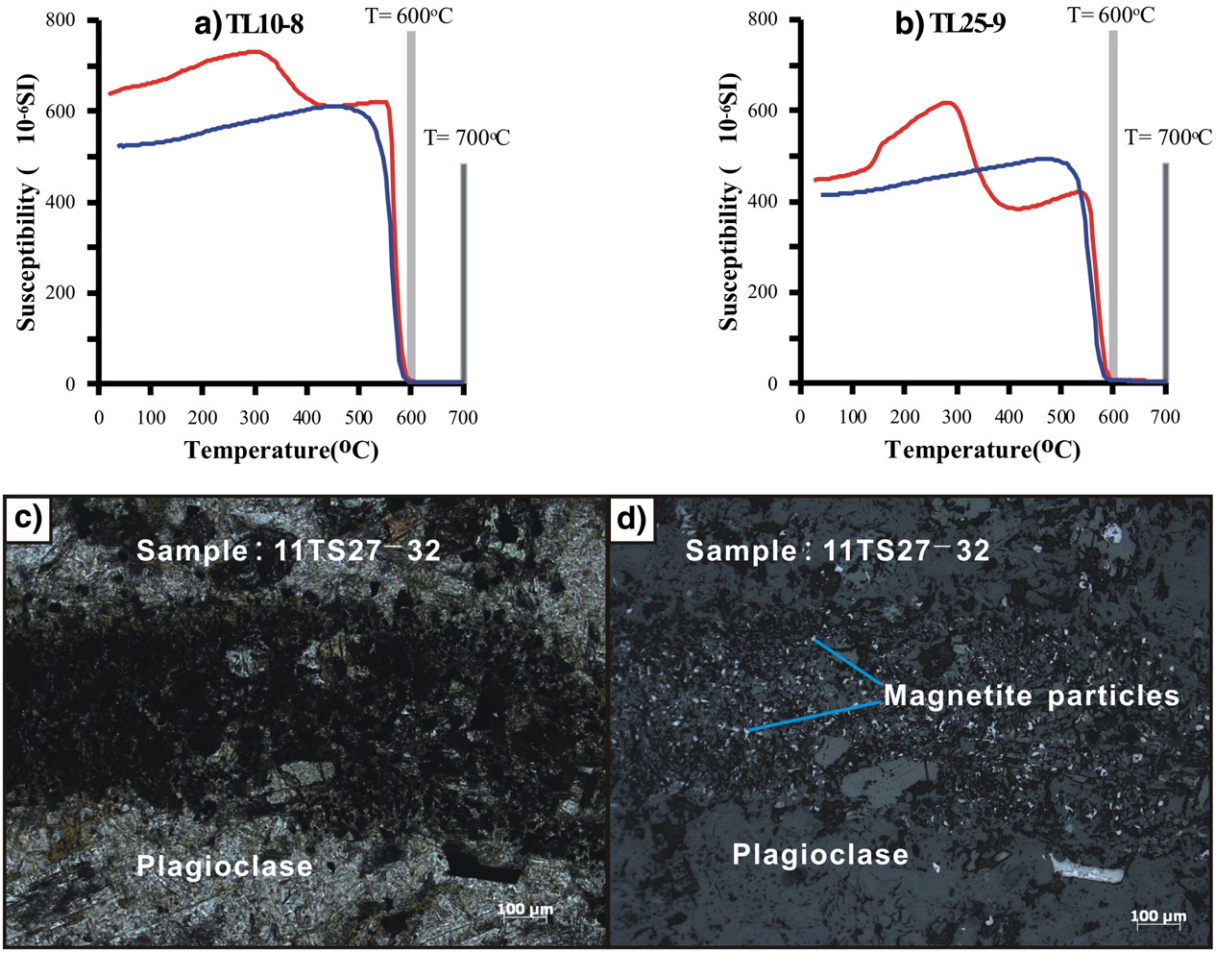


Fig. 4. Mineralogical investigations of the Duoni Formation volcanic rock samples. (a) and (b) Temperature dependence of the magnetic susceptibility during a heating–cooling cycle (room temperature to 700 °C for representative samples: (a) Sample TL10-8 is from the lower part of the volcanic emplacement (section 1, labeled in Fig. 1d). (b) Sample TL25-9 is from the upper part of the volcanic emplacement (section 1, labeled in Fig. 1d). Red lines show the heating curves in the air atmosphere, and blue lines show the cooling curves in the air atmosphere, decreasing from a maximum temperature of 700 °C to room temperature. (c) and (d) Photographs of polished thin-sections from a typical sample (sample 11TS27-32) from the volcanic rocks near Nagqu County: (c) Polarized-light photograph shows transparent minerals (bright areas), such as plagioclase and pyroxene. The abundant opaque minerals are magnetite particles. (d) Reflected-light photograph shows magnetite particles (bright spots) coexisting with common rock-forming minerals (*e.g.*, plagioclase and pyroxene).

5. Paleomagnetic methods and results

In the laboratory, we cut our paleomagnetic samples into standard 2.2-cm-long cylindrical specimens with diameters of 2.54 cm. The natural remanent magnetization (NRM) and subsequent remanent magnetizations were measured after each demagnetization step for all samples using a 2G Enterprises SQUID cryogenic magnetometer 755R at the Paleomagnetism and Geochronology Laboratory (PGL), Institute of Geology and Geophysics, Chinese Academy of Sciences, Beijing. However, the NRM intensity of the samples from several sites was beyond the measuring scope of the magnetometer. In these situations, we performed systematic demagnetization and remanence measurements using a JR-6 dual-speed spinning magnetometer at the Paleomagnetism and Environmagnetism Laboratory (PMEM) at China University of Geosciences in Beijing.

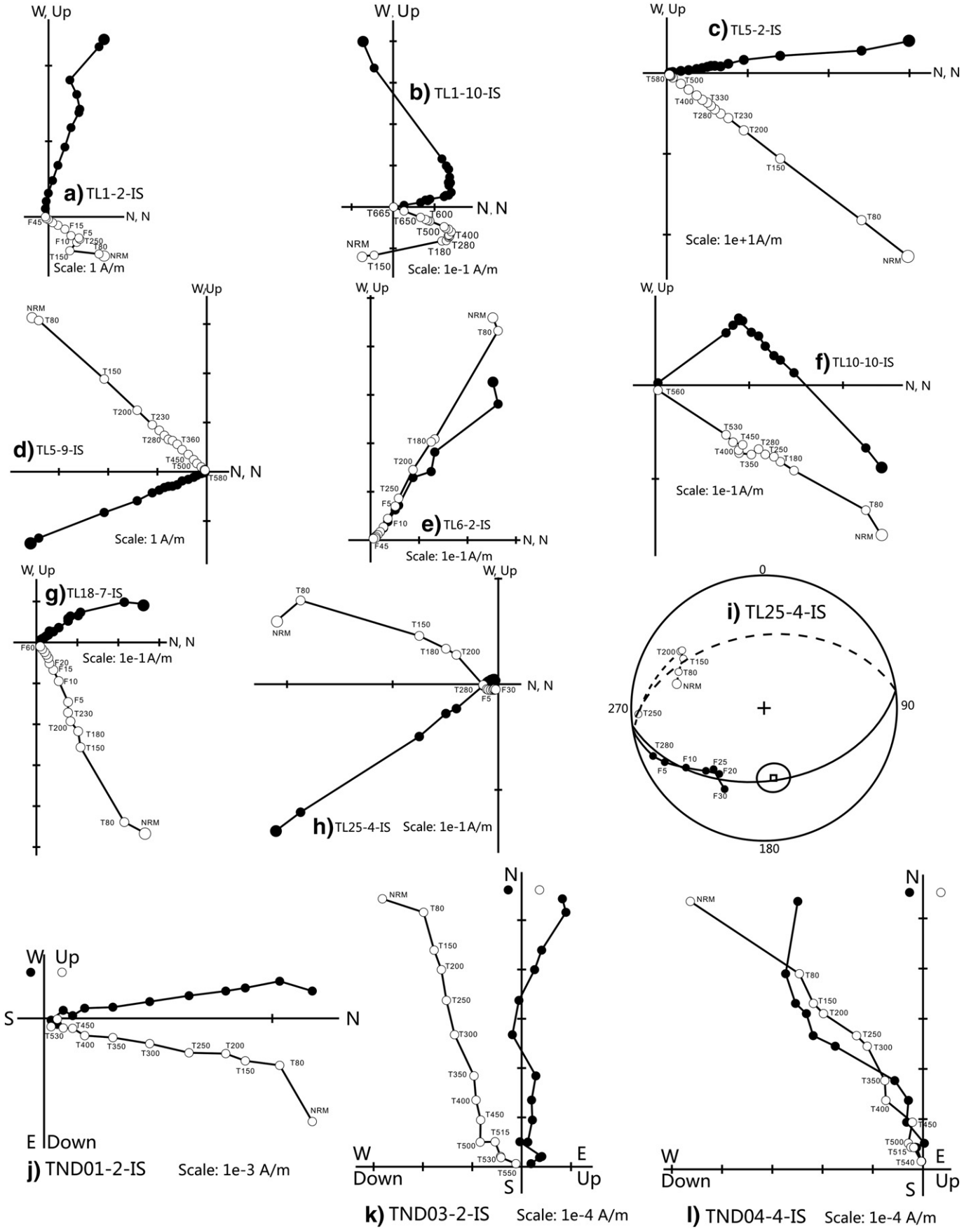
All thermal/AF demagnetization steps and subsequent remanence measurements were performed in a magnetically shielded room with a residual field of less than 300 nT. This method prevents the demagnetization and measurement procedures from being contaminated by the ambient geomagnetic field.

In the process of heating samples at the PGL and PMEM, we used an ASC TD-48 oven with internal residual fields of less than 10 nT. We designed our demagnetization scheme and steps according to the behavior observed in the thermomagnetic analysis. For one portion of the samples, we thermally demagnetized them in a stepwise manner until the maximum unblocking temperatures of 530–550 °C or 570–580 °C were approached (Fig. 5c, d, f). Because the dominant magnetic remanence carriers were magnetite or Ti-poor titanomagnetite particles in a portion of the samples, these samples were treated with a hybrid demagnetization technique to prevent chemical changes or mineralogical phase transformations. This technique involves an initial thermal demagnetization up to temperatures of 230–250 °C followed by AF demagnetization of 30–80 mT over the course of 18–22 steps (Fig. 5a, e, g). The results are described in detail below. The paleomagnetic software PMGSC (written by Randy Enkin) and the PaleoMac software pack (Cogné, 2003) were used for the stepwise demagnetization data analysis, statistical analysis and plotting of the data.

6. Paleomagnetic results

6.1. Paleomagnetic results of volcanic specimens

The initial NRM magnitudes of most of the volcanic samples are in the range of 10^{-4} A/m and 10^{-3} A/m. However, samples from the sites TL05, TL14, TL16, and TL19 exhibit NRM intensities that are 1–3 orders of magnitude higher (up to 10 A/m). All samples underwent stepwise thermal demagnetization from NRM up to 280–330 °C (Fig. 5a–i). Additionally, a portion of samples received increasing thermal


a) TL1-2-IS
Scale: 1 A/m
b) TL1-10-IS
Scale: 1e-1 A/m
c) TL5-2-IS
Scale: 1e+1A/m
d) TL5-9-IS
Scale: 1 A/m
e) TL6-2-IS
Scale: 1e-1A/m
f) TL10-10-IS
Scale: 1e-1A/m
g) TL18-7-IS
Scale: 1e-1A/m
h) TL25-4-IS Scale: 1e-1A/m
i) TL25-4-IS
j) TND01-2-IS Scale: 1e-3 A/m
k) TND03-2-IS Scale: 1e-4 A/m
l) TND04-4-IS Scale: 1e-4 A/m

demagnetization until the maximum unblocking temperatures of 530–550 °C or 570–580 °C were reached (Fig. 5c, d, f). Another portion of samples underwent subsequent AF demagnetization following thermal cleaning (*i.e.*, the hybrid demagnetization technique) (Fig. 5a, e, g, h). The AF steps of the hybrid technique feature initial increments as small as 5 mT that increase to 10 mT and 20 mT when the maximum magnetic field of 80–120 mT was reached (Fig. 5). The demagnetization results were evaluated on stereographic projections and orthogonal diagrams (Zijderveld, 1967).

Certain samples exhibit a single component between the first demagnetization step and the maximum unblocking temperatures (Fig. 5c, d, e). In general, most progressively demagnetized specimens exhibit relatively straightforward trajectories toward the origin of the orthogonal plots following the removal of a viscous or low-temperature component (LTC) by demagnetization steps at temperatures less than 250–300 °C. The LTCs are predominantly of normal polarity, except for three out of the 83 specimens, in which 4 or more consecutive temperature steps of the LTCs exhibited a reversed polarity. The LTCs cluster around the PGF and PDF directions based on *in situ* coordinates, and the directional grouping of the LTCs significantly deteriorates in the tilt-corrected coordinates.

Two components were isolated in the majority of the samples (Fig. 5a, b, f, g, h, i), and a high-temperature component (HTC) was primarily resolved across at least 4 consecutive temperature/alternating field steps from 400–450 °C to 570–580 °C and/or 660–670 °C. Principal component analysis (PCA) (Kirschvink, 1980) was applied to isolate distinct components with different unblocking temperatures, including the ChRMs isolations. For those specimens with different unblocking temperature intervals with much overlap, magnetic directions were fitted using remagnetization circles (Halls, 1978) and corresponding site-mean directions were calculated by using combined analysis technique (McFadden and McElhinny, 1988) (Fig. 5h, i; Fig. 6a, b; Table 1). The maximum angular deviations (MADs) (Kirschvink, 1980) are primarily less than 10°; only 5 volcanic specimen ChRMs are between 10°–15° (Table 1). Here, we defined HTCs as ChRM directions for all samples. Most ChRMs are of normal polarity, except two site-mean directions that exhibit a reversed polarity (Fig. 7e, f). Thus, both types of samples contain specimens that are north-seeking with a downward direction and south-seeking with an upward direction. The demagnetization behaviors conform to the thermomagnetic analysis, IRM acquisition, and backfield demagnetization results. These results indicate that both Ti-poor titanomagnetite and titanohematite are the dominant remanence carriers. Before performing statistical analysis to isolated ChRMs, we adopt the data filtering criteria described in detail below. Only ChRMs that met out data selection criteria were used to calculate the statistical mean direction. A rigorous filtering procedure was established because high-quality data is critical to the subsequent discussion of tectonic implications.

We calculated the statistical mean directions using the following procedure. We generally calculated site-mean directions from ChRMs that are fit by 4 or more consecutive demagnetization steps and that have MAD values of less than 15°. After this round of selection, 161 remaining ChRMs were used in the next step. To start performing an iterative process proposed by Vandamme (1994), firstly, each remaining ChRM was transformed into a corresponding VGP, and the Fisher mean VGP was calculated from this population. We then selected only VGPs that were within the optimum cutoff angle of this initial group mean VGP in tilt-corrected coordinates and then recalculated the group mean VGP. This procedure was continued until no outlier directions remained. In total, 6 iterative steps were required, and 39 ChRMs were rejected in this iterative process. The statistical site-mean directions were drawn from the remaining 122 ChRMs, and the remagnetization circles within each site and the resulting site-mean directions were calculated from 5 or more ChRM directions (and/or ChRM directions combined with remagnetization circles), although a few sites with 3–4 ChRMs that met our selection criteria were also used (Table 1). In contrast, sites with fewer than 3 ChRM directions or that feature only remagnetization circles contain no significantly statistical meaning. In these cases, the Fisher statistics are meaningless, and the site-mean directions were not calculated further (*i.e.*, samples TL13, 14, 16, 19, 20, and 21; labeled with superscript "c" in Table 1).

For several site-mean directions (*e.g.*, TL02, 03, 17, and 23; labeled with superscript "a") the ChRM directions exhibit north-seeking with gentle upward or south-seeking with gentle downward characteristics (Table 1). Although their ChRMs passed the iterative process, these anomalous site-mean directions are likely related to excursions of the geomagnetic field. Therefore, these sites were also discarded and were not used further in calculating the Fisher means. The ChRM directions for the site TL09 (labeled with the superscript "b" in Table 1) also have similar north-seeking with upward characteristics; however, the much steeper inclination of this site-mean direction distinguishes it from the site-mean directions discussed above. Characteristics of site TL09 can be regarded as a tectonic disturbance that occurred during a later stage, and this anomalous site-mean direction was removed from further calculation of the Fisher mean (Fisher, 1953).

In total, ChRMs from 14 sites and their associated 14 site-mean directions passed the filtering process. The statistical analysis yielded mean directions of Dg= 326.8°, Ig= 35.5°, kg = 24.0, α_{95g} = 8.3° and Ds= 356.3°, Is = 15.4°, ks= 22.5, α_{95}s= 8.6° before and after bedding correction, respectively, for the 14 sites (Fig. 7e,f; Table 1). The corresponding paleopole for the volcanic subgroup was calculated to be λ_p = 66.3°, φ_p = 281.1°, K = 23.6, A_{95} = 8.4° (N = 14) (Table 1).

6.2. Paleomagnetic results of sandstone specimens

In total, 41 samples from the 5 sandstone and siltstone sites near Nagqu County experienced progressive thermal demagnetization procedures from NRM up to the maximum unblocking temperatures of 525–530 °C or 550–560 °C (Fig. 5j,k,l). Certain samples display a single component between low-temperature steps and the maximum unblocking temperatures (Fig. 5). Similarly, most of the specimens exhibit straightforward trajectories toward the origin of the orthogonal plots following the removal of a viscous or low-temperature component by demagnetization steps at temperatures of 250–300 °C (Fig. 5j–l). Based on 3–4 temperature steps, the isolated LTCs were determined to be of normal polarity.

The HTC was primarily resolved over at least 4 consecutive temperature steps from 350–400 °C to 525–530 °C and/or 550–560 °C (Fig. 5j–l). Here, we define the HTC as the ChRM direction for all samples, similar to the procedure for the volcanic specimens.

All 41 specimens show north-seeking with downward directions. The demagnetization behaviors are consistent with the thermomagnetic analysis, IRM acquisition, and backfield demagnetization results, and this evidence indicates that Ti-rich titanomagnetite particles are the dominant remanence carriers.

All isolated ChRM directions were evaluated using principal component analysis (Kirschvink, 1980). The sandstone specimen data selection criteria and filtering processes were similar to those of the

Fig. 5. Orthogonal vector plots (a–l) (Zijderveld, 1967) for representative samples from the Lower Cretaceous volcanic rock specimens (a–i) and sandstone samples (j–l) of the Duoni Formation near Nagqu County. The directions are all plotted in geographic coordinates. The solid and open circles represent vector endpoints projected onto horizontal and vertical planes, respectively. Thermal (°C) and AF (mT) demagnetization steps are labeled with the capital characters T or F and are followed by a temperature (*e.g.*, T500) and peak demagnetization field (*e.g.*, F20), respectively. (h) and (i) Sample TL25-4: The orthogonal vector plot (h) cannot be fitted using the PCA method proposed by Kirschvink (1980), and the great circle technique is used to fit the demagnetization path (i).

volcanic specimens as performed before. For all 41 specimens, the site-mean directions were derived from ChRMs that were fit by 4 or more consecutive demagnetization steps and that featured MAD values of less than 10° (with more rigorous criteria than volcanic data). After selection using these two criteria, the 41 ChRMs were transformed to corresponding VGPs, and then performed iterative process as did for volcanic samples. Finally, the site-mean directions were calculated using 5 or more ChRM directions and $\alpha_{95} < 10°$ (Fig. 6c, d). In the ChRM direction filtering, only 2 ChRM directions were removed from further use in the calculation of Fisher site-mean directions and the statistical analysis (Table 1). In total, 39 ChRMs from 5 sites and their associated site-mean directions passed the filtering process. For the sandstone subgroup, the statistical analysis yielded mean directions of $D_g = 350.4°$, $I_g = 26.4°$, $k_g = 135.3$, $\alpha_{95g} = 6.6°$ and $D_s = 356.4°$, $I_s = 19.2°$, $k_s = 141.0$, $\alpha_{95s} = 6.5°$ before and after bedding correction, respectively (Fig. 7c, d; Table 1). The corresponding paleopole for the sandstone subgroup was calculated to be $\lambda_p = 68.4°$, $\varphi_p = 281.4°$, $K = 194.3$, $A_{95} = 5.5°$ ($N = 5$) (Table 1).

Combining these two directional data subgroups into one group yields an overall mean for all 19 sites of $D_g = 333.5°$, $I_g = 33.5°$, $k_g = 21.4$, $\alpha_{95g} = 7.4°$ and $D_s = 356.4°$, $I_s = 16.4°$, $k_s = 29.3$, $a_{95s} = 6.3°$ before and after bedding correction, respectively (Fig. 7g, h). The corresponding pole is $\lambda_p = 66.9°$, $\varphi_p = 281.2°$, $K = 31.4$, $A_{95} = 6.1°$ ($N = 19$) (Fig. 8a). When all the sandstone and volcanic site-mean directions are combined, the directional grouping was significantly higher in tilt-corrected coordinates than in *in-situ* coordinates (Fig. 7g, h). The results of the fold test (McFadden, 1990) are positive, with critical values (Xi_c) of 5.075 and 7.112 at the 95% and 99% probability levels, respectively. The parameter Xi_2 is equal to 9.397 and 3.820 in the *in situ* and tilt-corrected coordinates, respectively, suggesting a positive fold test at both the 95% and 99% confidence levels (Table 1). The precision parameter K reaches a maximum value when the strata are unfolded to a level of 79.1 ± 14.1% in the step-wise synfolding test (Watson and Enkin, 1993) (Table 1). These results also indicate that the ChRMs in the study area were acquired before folding activity occurred or during an very early folding stage.

6.3. Reliability of ChRM directions

The *in situ* mean direction of the LTC is consistent with both the PGF ($D = 0°$, $I = 48.3°$) and the PDF ($D = 0°$, $I = 50.6°$) at the sampling site (30.0°N, 91.1°E) (Fig. 7a,b). These results suggest that the LTC was originated from viscous overprinting during the Brunhes epoch and was influenced by the present geomagnetic field (Fig. 7a,b). This finding also supports the validity of the orientations of each core in the field.

The rock magnetic results from thermomagnetic analysis, IRM acquisition curves, and backfield demagnetization curves, together with petrographic investigations and the stepwise demagnetization behavior of the natural remanent magnetization, all indicate that the main carriers are Ti-poor titanomagnetite and Ti-poor titanohematite particles in the volcanic samples and Ti-rich titanomagnetite particles in the sandstone samples. The consistent paleodirections recorded by the different types of magnetic minerals for all samples suggest that the ChRM directions are consistent and primary.

Furthermore, the positive fold test (McFadden, 1990) and synfolding test (Table 1) for the combined volcanic and sandstone filtered ChRMs (*i.e.*, a total of 19 sites comprising 14 volcanic site-means and 5 sandstone site-means) show that the isolated ChRM directions were acquired prior to folding, as described in Section 6.2 (Table 1). The ChRMs were very likely acquired during the formation of the rocks before folding activity occurred and can therefore be considered essentially primary in origin.

Based on the geocentric axial dipole (GAD) model, another important factor for paleomagnetic research on volcanic rocks is whether the spot readings of an ancient geomagnetic field are time-averaged. This factor can be determined by calculating the VGP scatter parameter S_λ. The scatter of the virtual geomagnetic poles in this study was computed using the formula $S_\lambda = \{(1/N - 1) * \Sigma\theta_i^2\}^{1/2}$, where θ_i represents the angle between the ith VGP unit vector and the mean paleomagnetic pole unit vector, and N is the number of VGP unit vectors. We computed the VGP scatter to be 16.8° at the reference site of the 14 volcanic site-mean directions. This value falls within the interval of 16.6–21.0° in the latitudinal band of 0–20° during the period 110–195 Ma (McFadden et al., 1991), which indicates that the paleosecular variations have been adequately averaged out. Therefore, the paleomagnetic direction extracted from the samples can be regarded as the geocentric axial dipole field. Thus, the isolated paleomagnetic ChRM directions are suitable for tectonic discussions.

7. Discussion

The Cretaceous paleolatitude and paleogeographic reconstruction of the Lhasa block have been hotly debated over the past three decades (*e.g.*, the Lower Cretaceous Takena Formation red beds, the Woronggou lava flows, and the 130–110 Ma Zenong Group volcanics) (Pozzi et al., 1982; Westphal et al., 1983; Achache et al., 1984; Lin and Watts, 1988; Chen et al., 1993b; Sun et al., 2008; Tan et al., 2010; Chen et al., 2012; Sun et al., 2012; Ma et al., 2014; Yang et al., 2015). However, advances in Lower and Upper Cretaceous stratigraphy, structural geology analysis, paleontology, paleomagnetism and zircon U–Pb geochronologic dating (Sun et al., 2008; Zhu et al., 2011; Sun et al., 2012) have not yielded consistent results with regard to the paleolatitude of the Lhasa block during the Cretaceous. This debate in the geoscience community must be resolved.

To further constrain the paleolatitude of the Lhasa block and test whether inclination-shallowing occurred in the sandstone layers around the Nagqu area, 5 site-mean directions from sandstone layers were collected. Fortunately, the inclination-shallowing effect, which is pervasive across central Asia, is not present in this study. The site-mean directions of the sandstone specimens are consistent with those from the volcanic specimens (Fig. 7; Table 1). Thus, the inclination-shallowing effect, common in sandstone layers, does not need to be considered further in this study. Additional detailed discussion of inclination-shallowing is beyond the scope of this paper, and we refer interested readers to previously published paleomagnetic studies that explicitly focus on this issue (Cogné et al., 1999; Tan et al., 2003; Tauxe and Kent, 2004; Yan et al., 2005; Hankard et al., 2007; Dupont-Nivet et al., 2010a; Tan et al., 2010; Lippert et al., 2011; Huang et al., 2013).

7.1. Paleogeographic position of the Lhasa Block during the Early Cretaceous through Early Paleocene

Our geochronological results, together with previously published zircon dating results for the Duoni Formation volcano-sedimentary sequences (Liu et al., 2011), enable us to place a key constraint on certain controversial issues regarding the magmatic activity and emplacement ages of the extrusive magmas in the study area (Ciren and Xie, 2005; XIGS, 2005; Liu et al., 2011). Our results show that the peak magmatic activity occurred at 120 Ma (120.2 ± 0.5 Ma; Early Cretaceous), which is consistent with the results of Liu et al. (2011) from exposed volcanic flows in the same location (118–114 Ma, based on the dating results of two samples: 116.1 ± 0.4 Ma (MSWD = 1.4) and 115.9 ± 0.5 Ma (MSWD = 1.9)). Certain scholars have proposed that a large-scale magmatic event occurred between 130 and 110 Ma (Early Cretaceous) and extended along the northern Gangdese belt from Cuoqen and Shenza Counties in the west to Nagqu County in the east (Zhu et al., 2009; Ma and Yue, 2010; Liu et al., 2011).

Our paleomagnetic results are important for determining the paleogeographic position of the Lhasa block during the Early Cretaceous. From a geological perspective, significant northward convergence has occurred since the onset of the India–Asia collision (see Section 7.3). This convergence has been distributed between India, Tibet, and stable

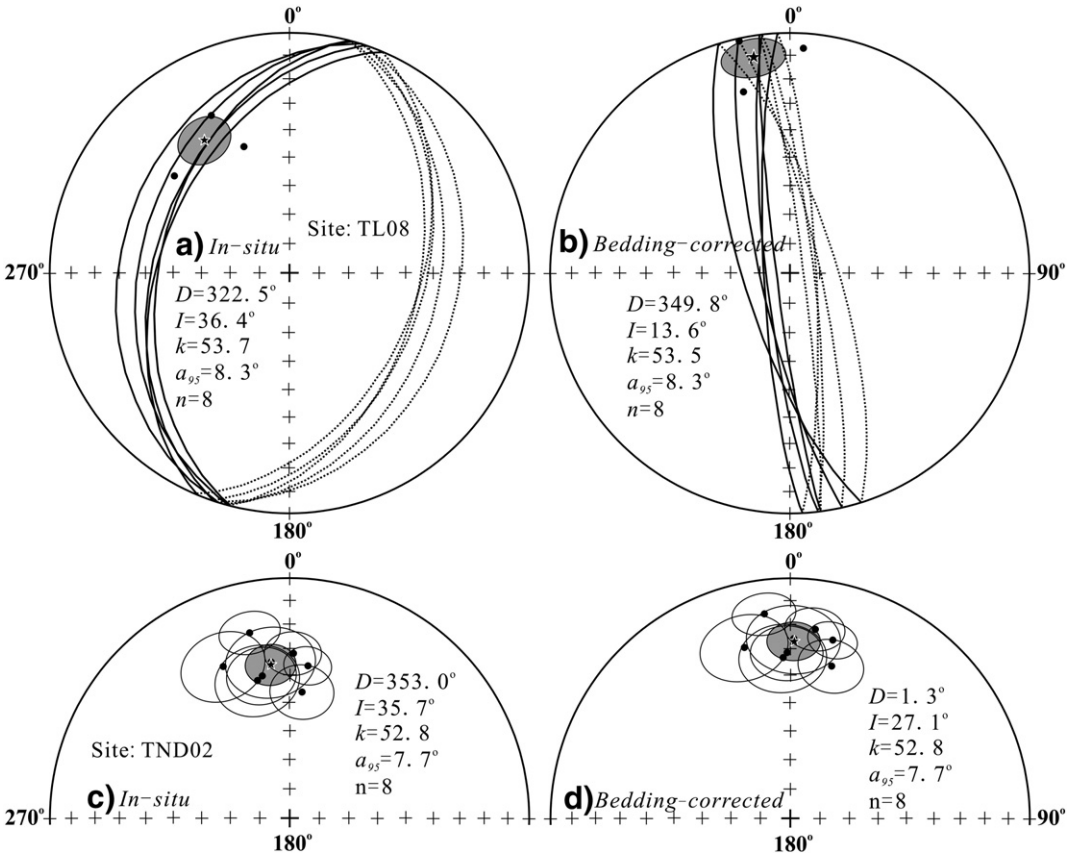
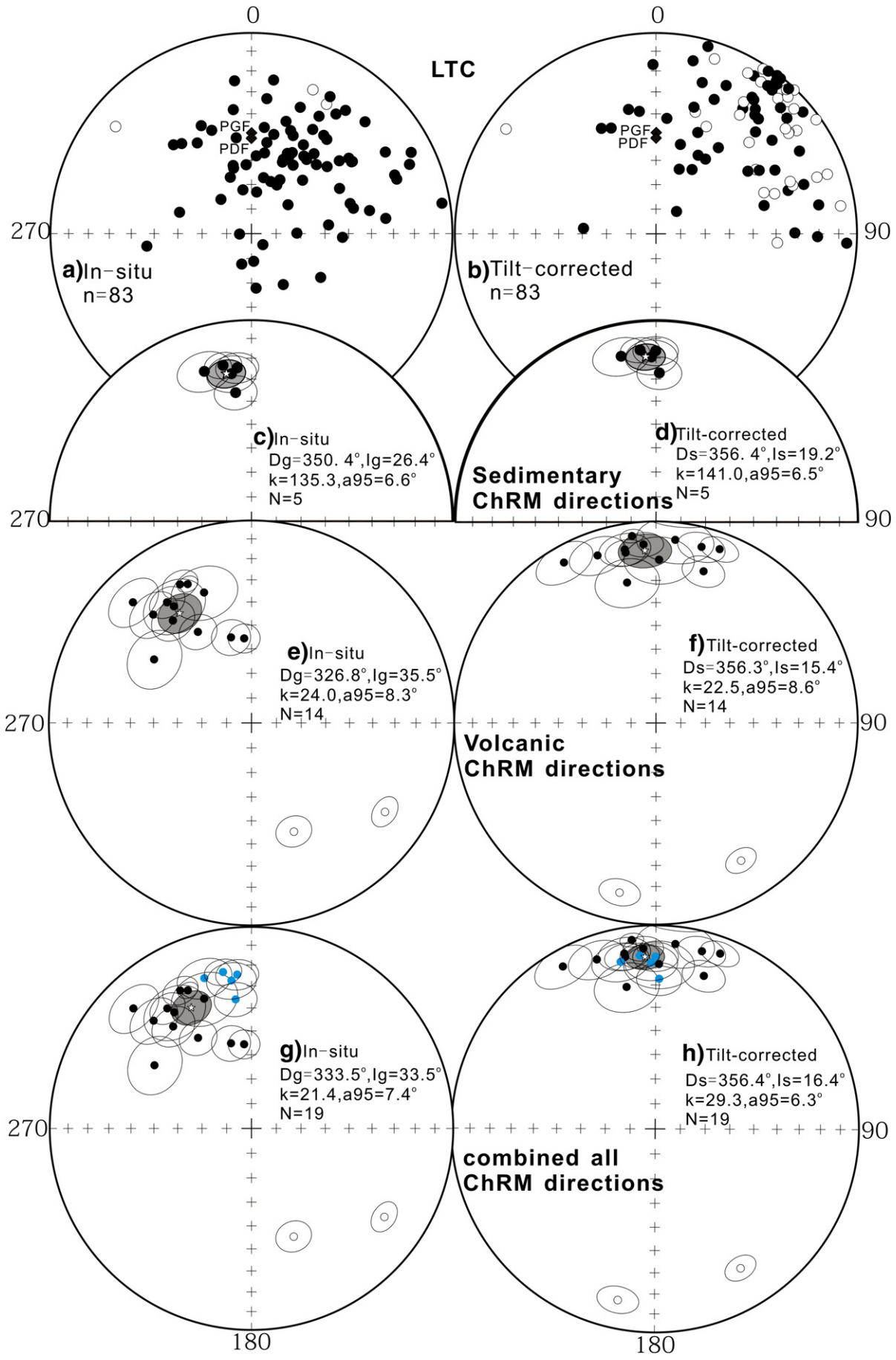


Fig. 6. Equal-area projections of the isolated ChRM directions and fitted great circles from volcanic samples from site TL08 before (a) and after (b) bedding corrections and the isolated ChRM directions from sandstone samples from site TND02 before (c) and after (d) bedding corrections. The corresponding statistical mean directions and corresponding 95% confidence ellipses are labeled as black stars with white margins and gray ellipses, respectively. All plots are projected onto the lower hemisphere. See text and Table 1 for details.

Asia, and different scholars argue for different amounts of continental shortening, *e.g.*, 188 ± 280 km, 620 ± 600 km, ~1500 km, or 2000 ± 550 km (Achache et al., 1984; Besse et al., 1984; Patriat and Achache, 1984; Chen et al., 2010; Liebke et al., 2010; Sun et al., 2010, 2012; Tang et al., 2013; Chen et al., 2014; Ma et al., 2014; Yang et al., 2015). This shortening is a consequence of the India–Asia collision and continued post-collisional northward movement of India relative to Asia (*i.e.*, continental subduction of the Indian plate beneath Asia to the north) (Aitchison and Ali, 2001; Aitchison et al., 2007; Ali and Aitchison, 2008; Tan et al., 2010; Chen et al., 2012; Meng et al., 2012; van Hinsbergen et al., 2012). Due to such intense and widely distributed deformation across Tibet and stable Asia, a well-constructed master APW path for the Eurasian continent (Besse and Courtillot, 2002; Schettino and Scotese, 2005; Torsvik et al., 2008, 2012; Cogné et al., 2013) cannot be used to predict the paleolatitudes of the Lhasa block during or after the India–Asia collision. Accordingly, the precise location of the southernmost edge of Asia (*i.e.*, the Lhasa block) prior to or during the initial contact of the two continents can only be reconstructed via paleomagnetic observations from the Lhasa block itself. This study's new paleopoles for the Lower Cretaceous Duoni Formation volcanosedimentary sequence in the Lhasa block indicate a paleolatitude of 7.2 ± 6.1°N for the reference site (30.0°N, 91.1°E) on the southernmost margin of the Lhasa block.

We have summarized the Cretaceous Lhasa block paleopoles collected in recent years using modern techniques (Table 2) and have projected them together with our paleopoles on an equal-area projection plot (Fig. 8a). In Table 2, we compiled the published Chinese and English results and evaluated them according to the seven-criterion system proposed by Van der Voo (1990), which assigned each result a quality factor Q-value. Although these selection criteria are rigorous, most of the poles have a Q-value of five or greater (Table 2). Many results failed to meet all seven criteria due to overlaps with the younger Cenozoic poles from the Lhasa block. For instance, the paleopole TLW meets five of the seven criteria: 1) well-determined rock age; 2) sufficient sample number ($n > 24$, $k \geq 10$, $\alpha_{95} < 16°$); 3) step-wise demagnetization; 4) robust fold test; and 5) structural control and tectonic coherence with the relevant craton. The same analysis was applied to the other poles.

In Table 2, each pole was evaluated based on its Q-value, according to the seven-criterion system. In this study, the poles were selected based on the following two factors: 1) a Q-value equal to or great than 5 (criteria 1, 2, 4, and 5 are particularly important to our subsequent discussion, and a pole with a 5 or greater Q-value would still be rejected if it fails to meet any one of these criteria) and 2) field tests (especially the fold test) are available (these tests are necessary for selecting poles with high-quality pole data, and poles that do not satisfy criterion 4 would not be considered). Based on these data selection criteria, all poles listed in Table 2, except pole NL, have a Q-value of 5 or greater; NL was excluded from further analysis due to its low Q-value (Q = 3) (Table 2). Additionally, despite its Q-value of 5, pole WV was also not considered in this study because it fails to meet criterion 4 (no fold test). And this study produced pole NV, which has a high Q-value of 6 and passed the fold test. In summary, all poles, except NL and WV, were used in our analysis, and the small-circle analytical method (Mardia and Gadsden, 1977) was used to fit the filtered 13 poles (*i.e.*, poles SR, TP, TNA, TaS, TLW, TC, SM, ZL, YV, CV, CQV, CQR, and NV). The fitting circle passes through the 13 selected poles with a colatitude of 76.9 ± 2.7° centered on the reference site (30.0°N, 91.1°E). Thus, through conversion, the corresponding paleolatitude for the reference site was found to be 13.1 ± 2.7°N (Fig. 8a; Table 2).


LTC
PGF
PDF
a)In-situ
n=83
b)Tilt-corrected
n=83
c)In-situ
Dg=350. 4°,Ig=26.4°
k=135.3,a95=6.6°
N=5
d)Tilt-corrected
Ds=356. 4°,Is=19.2°
k=141.0,a95=6.5°
N=5
Sedimentary
ChRM directions
e)In-situ
Dg=326.8°,Ig=35.5°
k=24.0,a95=8.3°
N=14
f)Tilt-corrected
Ds=356.3°,Is=15.4°
k=22.5,a95=8.6°
N=14
Volcanic
ChRM directions
g)In-situ
Dg=333.5°,Ig=33.5°
k=21.4,a95=7.4°
N=19
h)Tilt-corrected
Ds=356.4°,Is=16.4°
k=29.3,a95=6.3°
N=19
combined all
ChRM directions
0
90
180
270

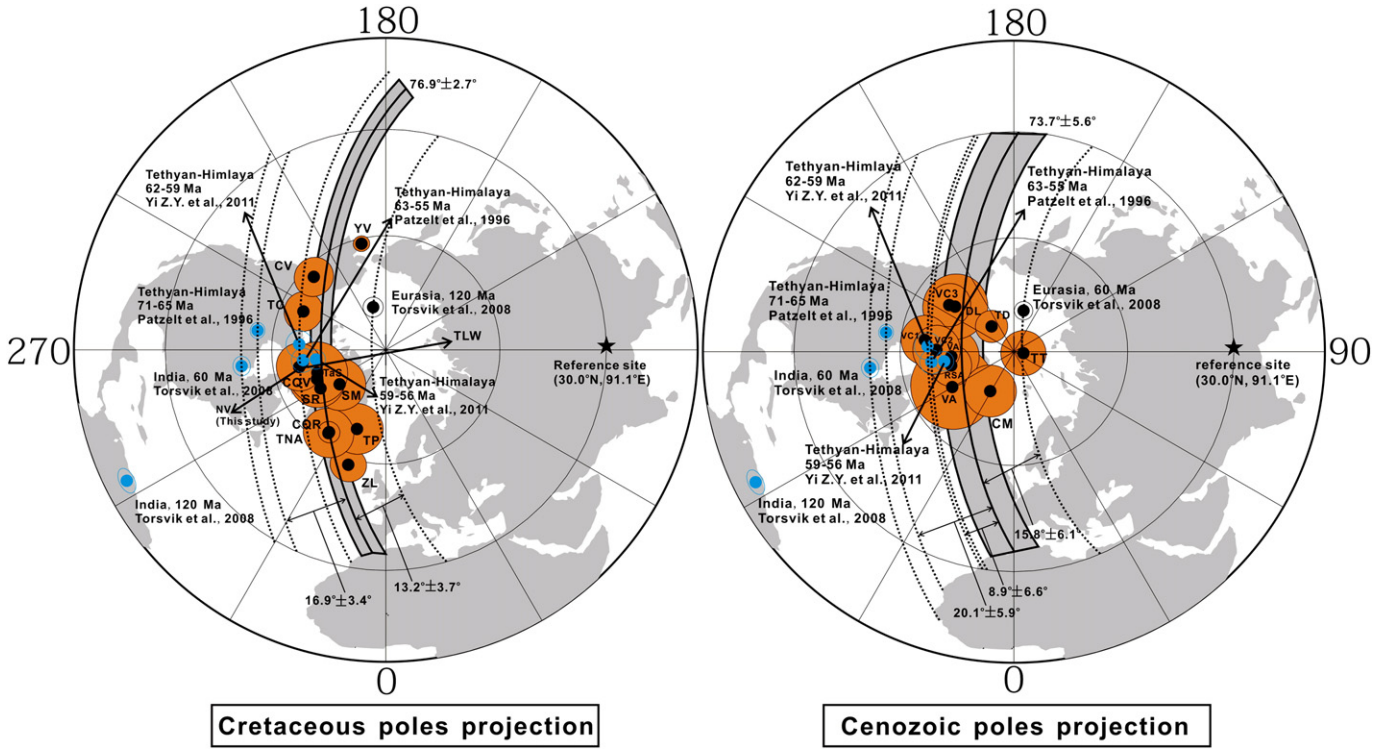


Fig. 8. Equal-area projections showing the distribution of Cretaceous paleopoles (a) and Cenozoic paleopoles (b) of the Lhasa block and adjacent blocks based on this study and previously published results, with corresponding 95% confidence circles. See Table 2 for values and sources. The 120 Ma and 60 Ma APW reference poles for the Eurasian continent and Indian shield (Torsvik et al., 2008) are also projected onto the Northern Hemisphere. The black star indicates the sampling location, which was selected as the reference site (30.0°N, 91.1°E) when calculating the paleolatitude of the Lhasa block in this study. The dotted line represents the small-circle running through the reference poles of Eurasia, India and the Tethyan–Himalayan block centered on the reference site. Black dots with the orange-filled circles represent the selected poles that were fit by the small-circle method and the 95% corresponding confidence circle. The blue dots with confidence circles represent the poles with corresponding confidence circles for the Indian and Tethyan–Himalayan blocks. The value 13.2 ± 3.7° is the latitudinal displacement and corresponding confidence level. The light gray strip area with solid black line in (a) and (b) represents the small-circle and its 95% confidence limit passing through the paleopoles centered on the reference site (30.0°N, 91.1°E) in this study. Abbreviations are drawn from following references: Cretaceous paleopoles (NL, SR, TP, TNA, TaS, TLW, TC, TC, SM, WV, ZL, YV, CV, CQV, CQR, and NV): 1. Lin and Watts, 1988; 2. Tan et al., 2010; 3. Pozzi et al., 1982; 4. Achache et al., 1984; 5. Chen et al., 1993a,b; 6. Sun et al., 2012; 7. Sun et al., 2008; 8. Chen et al., 2012; 9. Ma et al., 2014; 10. Tang et al., 2013; 10+, Yang et al., 2015. Cenozoic poles (VA, TD, TT, VC1, VC2, VC3, RS, RSA, DL, and CM): 11. Achache et al., 1984; 12. Dupont-Nivet et al., 2010a; 13. Dupont-Nivet et al., 2010b; 14. Tan et al., 2010; 15. Chen et al., 2010; 16. Sun et al., 2010; 17. Liebke et al., 2010; 18. Meng et al., 2012. Tethyan–Himalayan and Indian Shield poles: 19. Patzelt et al., 1996; 20. Yi et al., 2011; 21. Torsvik et al., 2008; 22. Chen et al., 2014. See the text and Table 2 for details.

The same analysis is also applied to the Cenozoic paleopoles (Fig. 8b). Each pole was evaluated based on its Q-value, according to the necessary information in the literature. All poles have a 5 and/or greater Q-value, satisfied criteria 1, 2, 4, and 5, and passed the fold test. Therefore, no poles were excluded from further analysis. In an equal-area projection plot (Fig. 8b), these poles exhibit a similar small-circle distribution but are more scattered than the Cretaceous poles, implying that relative rotations occurred between the different study areas in the Lhasa block. The small-circle method was also used to fit 10 selected poles (*i.e.*, VA, TD, TT, VC1, VC2, VC3, RS, RSA, DL, and CM) centered on the reference site in this study with a best-fitting colatitude of 73.7 ± 5.6°, suggesting a corresponding latitude of 16.3 ± 5.6°N centered on the reference site. In comparison with the fitted result for the Cretaceous poles (13.1 ± 2.7°N), this result clearly indicates that the Lhasa block experienced little latitudinal convergence (3.2 ± 6.2°) between the Early Cretaceous (120 Ma) and the Early Paleocene, *i.e.*, prior to the India–Asia collision. Therefore, the Lhasa block experienced no paleomagnetically detectable northward movement between the Early Cretaceous and the Early Paleocene and remained essentially stationary at ~13°N during this period. In fact, a number of paleomagnetists have noted that the Lhasa block likely remained at a latitude of ~10–15°N between the Cretaceous and early Paleogene before the India–Asia collision, based on a series of paleomagnetic investigations in the Lhasa block (Achache et al., 1984; Sun et al., 2008; Chen et al., 2010; Liebke et al., 2010; Sun et al., 2010, 2012; Chen et al., 2014; Yang et al., 2015).

During the past two decades, several paleomagnetic studies from the Takena Formation and other rock suites have yielded consistently low latitudes in the range of ~10–15°N (*e.g.*, 9.4 ± 4.9°N (Lin and Watts, 1988), 13.5 ± 5.4°N (Sun et al., 2010), and 13.2 ± 7.9°N (Achache et al., 1984)) during the Cretaceous. The Lhasa block likely stayed at a low-middle latitude of 10–15°N and exhibited little northward motion during the time interval between the deposition of the Takena Formation red beds during Albian–Aptian times (125–100 Ma) and the formation of the Linzizong Group (68–43 Ma). Assuming that the onset of the India–Asia collision occurred no later than 50 Ma (see Section 7.2), the quiescence of the Lhasa block during the formation of the Duoni Formation in the northern part of the Lhasa block and the formation of the

Fig. 7. (a, b) Equal-area projections of the low-temperature components (LTCs) isolated from 83 samples, and (c, d) equal-area projections of the site-mean ChRM directions with corresponding 95% confidence ellipses before and after tilt correction (left and right sides) for 5 sandstone sites near Luoma town in the Nagqu area. (e, f) Equal-area projections of 14 filtered site-mean characteristic remanent magnetizations isolated from volcanic rocks with corresponding 95% confidence ellipses before and after tilt correction (left and right sides) for the locality near Luoma town in the Nagqu area. (g, h) Equal-area projections for all 19 selected site-mean characteristic remanent magnetizations isolated from the volcanic and sandstone rocks (blue dots) with corresponding 95% confidence ellipses before and after tilt correction (left and right sides). Solid and open symbols represent projections onto lower and upper hemisphere, respectively. Stars with light gray-colored ellipses indicate the overall mean directions and the 95% confidence limits before and after tilt correction, respectively. Black diamonds labeled in (a, b) are equal projections of the PGF and PDF directions calculated for the reference site.

Table 2
Summary of Paleomagnetic poles published in Chinese and English literature over the past two decades. This study's results from the Duoni Fm. (northern part of the Lhasa block) and paleopoles from the Tethyan–Himalayan block and Indian Shield are also listed below. See Fig. 8a, b.

Abbr.	Age (Ma)	Formation	λs (°N)	φs (°E)	N	λp (°N)	φp (°E)	A95 (dp/dm)(°)	Criteria (Q)	Plat (°N)	Sources
	Reference site (30.0°N, 91.1°E) *Cretaceous Paleopoles for the Lhasa block*		Sampling site			Cretaceous paleopoles					
NL	95–100 (Ar/Ar)	Nagqu lava	31.5	92.0	9	78	282	4.0/6.9	123□□□□(3)	18.2 ± 4.0	1
SR	Albian–Late Cretaceous	Shexing Fm., red beds, Linzhou	29.9	91.2	43	70.2	300.5	1.9	123F5D□(6)	12.4 ± 1.9	2
TP	Albian–Aptian	Takena Fm. red beds, Linzhou	29.4	91.0	7	68	340	6.7/11.6	123 F5□7(6)	20.3 ± 6.7	3
TNA	Albian–Aptian	Takena Fm. red beds, northern region	30.0	91.0	6	63.5	325.4	6.5	123 F5□□(5)	12.8 ± 6.5	4
TaS	Albian–Aptian	Takena Fm. red beds, southern region	30.0	91.0	8	71.2	288.4	7.9	123 F5□□(5)	11.9 ± 7.9	4
TLW	Albian–Aptian	Takena Fm. red beds, Linzhou	29.9	91.2	8	68	279	3.5/6.9	123 F5□□(5)	8.2 ± 3.5	1
TC	Albian–Aptian	Combined pole of Late Cretaceous, sandstone and limestone, western Tibet	33.7	80.4	14	66.2	245	5.1	123F5D7(7)	8.3 ± 5.1	5
SM	Late Cretaceous	Shexing Fm., Maxiang,	29.9	90.7	20	75	306.7	6.8	123F5D□7(6)	17.5 ± 6.8	6
WV	114.2 ± 1.1 Ma, SHRIMP	Woronggou Fm., volcanic rocks, Deqing	30.5	90.1	15	66.4	220.3	6.9	123□5□7(5)	13.8 ± 6.9	7
ZL	110–130 Ma, U–Pb dating	Zenong Gr., lava, Cuoqen	31.3	85	18	58.2	341.9	4.6	123F5R7(7)	16.0 ± 4.6	8
YV	132–120 Ma, U–Pb dating	Yanhu area, volcanic rocks	32.34	82.57	51	61.4	192.9	2.1	123 F5□□(5)	20.7 ± 2.1	9
CV	99–93 Ma, (Ar/Ar)	Cuoqin area, volcanic rocks	30.96	85.2	18	63.1	224.6	5.1	123 F5□□(5)	10.1 ± 5.1	10
CQV	121–117 Ma, U–Pb dating	Cuoqin area, Dianzhong Fm., volcanics	31.12	84.37	12	70.5	292.9	7.4	123 F5□□(5)	11.7 ± 7.4	10+
CQR	99.6–65.5	Cuoqin area, Jingzhushan Fm., red beds	31.15	84.86	33	63.8	325.4	2.8	123 F5□□(5)	13.0 ± 2.8	Recalculated. from 10+
NV	120.2 ± 0.5 Ma, U–Pb dating	Nagqv, volcanic rocks	31.3	91.9	19	66.9	281.2	6.1	123F5D□(6)	7.2 ± 6.1	This study
Abbr.	Age (Ma)	Formation	λs (°N)	φs (°E)	N (n)	λp (°N)	φp (°E)	A95 (°)	Criteria (Q)	Plat (°N)	Sources
	Reference site (30.0°N, 91.1°E) *Cenozoic Paleopoles for the Lhasa block*		Sampling site			Cenozoic paleopoles					
VA	65–45	Volcanic rocks	30.0	91.0	8	71.4	299.8	11.0	123F5D□(6)	13.4 ± 11.0	11
TD	54–47	Felsic tuff	30.0	91.1	37	81.2	221.4	4.2	123F5D□(6)	24.1 ± 4.2	12, 13
TT	43–40	Tuffs	30.0	91.2	9	87.5	81.1	5.9	123 F5□□(5)	32.5 ± 5.9	2
VC1	64–60	Dianzhong Fm. volcanic rocks	30.0	91.1	20	66.4	262.5	6.3	123F5D□(6)	6.6 ± 6.3	15, 22
VC2	60–50	Nianbo Fm. volcanic rocks	30.0	91.1	13	69.7	268.6	6.3	123F5D□(6)	9.7 ± 6.3	15, 22
VC3	50–44	Pana Fm. volcanic rocks	30.0	91.1	18	69.1	234.2	5.6	123F5D□(6)	12.7 ± 5.6	15, 22
RS	~55, Ar/Ar	Rhyolitic tuffs	30.1	90.9	14	73.6	274.3	7.3	123F5D□(6)	13.6 ± 7.3	16
RSA	~55, Ar/Ar	Combined pole	30.1	90.9	22	73.2	282.4	5.4	123F5R□(6)	13.5 ± 5.4	4, 16
DL	~53	Dykes	30.1	91.0	10	70.6	232.5	8.5	123F5R□(6)	14.3 ± 8.5	17
CM	E1–E2	Upper Cuojiangding Gr.	29.9	84.3	(62)	78.0	329.0	6.8	123 F5□□ (5)	23.2 ± 6.8	18
Abbr.	Age (Ma)	Formation	λs (°N)	φs (°E)	N/(n)	λp (°N)	φp (°E)	A95 (dp/dm) (°)	Criteria (Q)	Plat (°N)	Sources
	Reference site (30.0°N, 91.1°E) K2/Pg1 Paleopoles for the Himalayan block and Indian shield		Sampling site			Himalayan poles					
Tethyan Himalaya											
TPK2	71–65	Zongshan Fm., limestone, TH	28.3	88.5	(156)	55.8	261.4	4.4/8.6	123 F5□7 (6)	(−)3.8 ± 4.4	19
TPE1	63–55	Zongpu Fm., limestone, TH	28.3	88.5	(113)	65.4	277.6	3.8/7.6	123□5□7 (5)	5.5 ± 3.8	19
TYE1-1	62–59	Zongpu Fm., marine sediments, Gamba	28.3	88.5	18 (189)	67.3	266.3	3.5	123 F5□7 (6)	7.4 ± 3.5	20
TYE1–2	59–56	Zongpu Fm., marine sediments, Gamba	28.3	88.5	14 (141)	71.6	277.8	2.5	123 F5□7 (6)	11.7 ± 2.5	20
Indian Shield											
In120	120	APW reference pole	/	/	/	7.7	296.8	2.6	/	(−)44.9 ± 2.6	21
In60	60	APW reference pole	/	/	/	51.6	276.4	2.4	/	(−)8.3 ± 2.4	21

Abbreviations are as below: Abbr., abbreviations of paleopoles discussed in the text; λs (°N), φs (°E): latitude and longitude of different studies' sampling sites; N/(n): number of sites or specimens used to calculate the Fisher site-mean directions; λp (°N), φp (°E): latitude and longitude of previously published paleopoles and the results from this study; A_{95}/[dp/dm]: half-angle of the 95% confidence cone of the pole; Criteria (Q): data quality evaluation and quality factor (criteria numbers met) after Van der voo (1990) [1. Well-determined rock age; 2. Sufficient sample numbers, n ≥ 24, k (or K) ≥10,α_{95} ≤ 16°; 3. Systematic step-wise demagnetization; 4. Field tests; 5. Structural control and tectonic coherence with the relevant craton; 6. Presence of reversal; 7. No resemblance to paleopole with younger age. The numbers indicate that the data satisfied these criteria. For criterion 4, F stands for positive fold test; D, presence of dual polarity; and R, positive reversals test. The symbol □ indicates failure to meet this criterion.]. Plat (° N), paleolatitude computed for the reference point at 30.0°N, 91.1°E.
Sources code, 1. Lin and Watts, 1988; 2. Tan et al., 2010; 3. Pozzi et al., 1982; 4. Achache et al., 1984; 5. Chen et al., 1993a,b; 6. Sun et al., 2012; 7. Sun et al., 2008; 8. Chen et al., 2012; 9. Ma et al., 2014; 10. Tang et al., 2013; 10+. Yang et al., 2015; 11. Achache et al., 1984; 12. Dupont-Nivet et al., 2010a; 13. Dupont-Nivet et al., 2010b; 14. Tan et al., 2010; 15. Chen et al., 2010; 16. Sun et al., 2010; 17. Liebke et al., 2010; 18. Meng et al., 2012; 19. Patzelt et al., 1996; 20. Yi et al., 2011; 21. Torsvik et al., 2008; 22. Chen et al., 2014.

Linzizong Group volcanics further supports the subduction of Indian lithosphere beneath southern Asia (the Lhasa block). This hypothesis has also been suggested by geophysical studies of crustal and lithospheric structure beneath the Himalayan–Tibetan orogeny (Powell and Conaghan, 1973; Zhao et al., 1993; Owens and Zandt, 1997; Kind, 2002; Zhao et al., 2010, 2011, 2011).

7.2. Implications for timing of the India–Asia collision

Before beginning our analysis on the timing of the India–Asia collision, we should assess the accuracy of the paleolatitude of Asia's southernmost margin (*i.e.*, the southern margin of the Lhasa block) prior to India–Asia collision. Therefore, we compare our results with the paleolatitude of the Tethyan–Himalayan block, which is often taken to represent the northernmost margin of Greater India (Patzelt et al., 1996; Tong et al., 2008; Yi et al., 2011; Huang et al., 2015a).

Fig. 8 shows the equal-area projection of paleopoles from APW reference poles for the Indian shield and Eurasian continent (120 and 60 Ma poles are used for both continents). Researchers have constructed a synthetic APW path for the Eurasian continent since 200 Ma via filtering and selection from a paleomagnetic database (Besse and Courtillot, 2002; Schettino and Scotese, 2005; Torsvik et al., 2008, 2012; Cogné et al., 2013). New APW reference poles were recently published, and these newly updated reference paleopoles were adopted in this study (Torsvik et al., 2008).

Additionally, due to the large-scale deformation and magnitude of convergence within the Tethyan–Himalaya, cratonic India, Tibet, and stable Asia since the initiation of the India–Asia collision, the reference APWP curves for India and Eurasia cannot accurately predict the paleopositions of the southernmost margin of Asia or the northernmost margin of Greater India. To directly compare the paleolatitudes of the two continents, our results have been combined with previously published paleomagnetic results from the southern margin of the Lhasa block and the Tethyan–Himalayan block, which represents the northern margin of Greater India, thereby spanning both sides of the IYZSZ.

On the Indian side, the paleolatitude of the northern Greater Indian margin was calculated using data collected directly south of the IYZSZ. Data from the Zongshan Formation (71–65 Ma) ($\lambda p = 55.8°$, $\varphi p = 261.4°$ with dp/dm = 4.4°/8.6°) and Zongpu Formation (63–55 Ma) ($\lambda p = 65.4°$, $\varphi p = 277.6°$ with dp/dm = 3.8°/7.6°) were collected from the Gamba and Duela areas, respectively (Patzelt et al., 1996) (Table 2; Fig. 8). The latest paleomagnetic study on the Zongpu Formation in the Gamba area divided the formation into two periods: the period 62–59 Ma with $\lambda p = 67.3°$, $\varphi p = 266.3°$ and $A_{95} = 3.5°$ and the period 59–56 Ma with $\lambda p = 71.6°$, $\varphi p = 277.8°$ and $A_{95} = 2.5°$ (Yi et al., 2011) (Table 2; Fig. 8). The reference site (30.0°N, 91.1°E) of the northern margin of the Tethyan–Himalayan block was therefore located at 3.8 ± 4.4°S at 68 ± 3 Ma (Zongshan Formation; 71–65 Ma) and 5.5 ± 3.8°N at 59 ± 4 Ma (Zongpu Formation; 63–55 Ma) based on the results of Patzelt et al. (1996). The data from Yi et al. (2011) yields paleolatitudes of 7.4 ± 3.5°N ($\lambda p = 67.3°$, $\varphi p = 266.3°$ with $A_{95} = 3.5°$) and 11.7 ± 2.5°N ($\lambda p = 71.6°$, $\varphi p = 277.8°$ with $A_{95} = 2.5°$) for the time periods 62–59 Ma (60.5 ± 1.5 Ma) and 59–56 Ma (57.5 ± 1.5 Ma), respectively.

In contrast, on the Asian side, the Cretaceous and lower Cenozoic results of previous Lhasa block studies are hotly debated and highly controversial with regard to the timing of initial contact between India and Asia in the Chinese and English literature (Pozzi et al., 1982; Westphal et al., 1983; Achache et al., 1984; Lin and Watts, 1988; Chen et al., 1993a,b; Sun et al., 2008; Tan et al., 2010; Chen et al., 2012; Meng et al., 2012; Sun et al., 2012; Huang et al., 2013; Tang et al., 2013; Lippert et al., 2014; Ma et al., 2014; Yang et al., 2015; Huang et al., 2015b,c,d). Combining our data with previously published results, we calculated comparable paleolatitudes of 13.1 ± 2.7°N and 16.3 ± 5.6°N for the southernmost margin of the Lhasa block (reference site: 30.0°N, 91.1°E) during the Early Cretaceous and Early Paleocene, respectively, using the small-circle method. Here, we do not discuss the timing of initial contact between India and Asia based on the small-circle method for Cenozoic poles for two reasons: 1) In the Cenozoic small-circle fitting results, certain poles with younger ages (50–40 Ma) were included. Although the dating results for the volcanics are accurate, these volcanic rocks were likely post-collisional products, and their inclusion in the analysis would produce an artificially younger age for the timing of the initiation of India–Asia collision. 2) Because the Lhasa block experienced only insignificant latitudinal convergence of 3.2 ± 6.2° between the Early Cretaceous (~120 Ma) and the early Cenozoic prior to the India–Asia collision, no paleomagnetically detectable latitudinal movement is identifiable within error. Therefore, we conclude that the small-circle analysis results for the Cretaceous poles (13.1 ± 2.7°N) can be reasonably used to discuss the Paleolatitude of the Lhasa block and subsequently the timing of India–Asia collision. This paleolatitude is essentially identical to the paleolatitude of the Tethyan–Himalayan block (*i.e.*, the northern Greater Indian margin) at the 95% probability level, which indicates that the initial contact between India and Asia likely occurred no earlier than 62–59 Ma but no later than 59–56 Ma based on the results of Yi et al., 2011.

Furthermore, a large number of paleomagnetic studies have advocated the view that the Lhasa block remained essentially stationary between the Early Cretaceous and the Middle Eocene (Achache et al., 1984, 1991; Chen et al., 2010; Huang et al., 2010; Sun et al., 2010; Chen et al., 2012; Sun et al., 2012; Tang et al., 2013; Chen et al., 2014; Lippert et al., 2014). Across the suture zone, the paleolatitudes for the northernmost Greater Indian margin progressively increase northward with decreasing age, ranging from 3.8 ± 2.0°S at 68 ± 3 Ma to 7.4 ± 3.5°N at 60.5 ± 1.5 Ma to 11.7 ± 2.5°N at 57.5 ± 1.5 Ma, based on the combined results of Patzelt et al. (1996) and Yi et al. (2011). Therefore, a significant gap (16.9 ± 3.4°) clearly existed between the northernmost margin of the Tethyan–Himalaya and the southernmost margin of the Lhasa block at 68 ± 3.0 Ma, and the gap rapidly decreased to 1.4 ± 3.7° by 57.5 ± 1.5 Ma. This pattern indicates that 1) the northernmost margin of the Tethyan–Himalaya was located farther south (16.9 ± 3.4°, or 1860 ± 370 km) at 68 ± 3 Ma and 2) the northernmost margin of the Tethyan–Himalaya had reached the southernmost margin of the Lhasa block (1.4 ± 3.7°, or 150 ± 400 km) by 57.5 ± 1.5 Ma. Based on this reasonable deduction, the initial contact between India and Asia must have occurred before 57.5 ± 1.5 Ma.

Assuming that the northernmost Greater Indian margin was at 7.4 ± 3.5°N during 62–59 Ma based on paleomagnetic evidence (Yi et al., 2011), and that the Indian shield was drifting northward at a velocity of ~18–19.5 cm/yr at around 60.5 ± 1.5 Ma (Klootwijk et al., 1992), we calculate that the paleolatitude of the northern margin of Greater India intersected with the paleolatitude of the southern margin of the Lhasa block at the 95% probability level approximately 1.2 Ma later (*i.e.*, 59.3 Ma). In other words, their paleolatitudes overlapped at ca. 59.3 Ma at a latitude of 13.1 ± 3.5°N, which is indicative of the onset of the Indo-Asia collision. This conclusion is based on the assumption that the velocity of the Indian shield was at the lower end (18 cm/yr) of its velocity range (18–19.5 cm/yr). In reality, the Indian continent likely drifted at a speed somewhat faster than this value (*e.g.*, 19 cm/yr). Therefore, the age of initial contact of between India and Asia should be earlier than our conservative estimate.

The hypothesis that the onset of India–Asia collision occurred at ca. 59.3 Ma is supported by substantial geological evidence (Beck et al., 1995; Zhang et al., 2001; Mo et al., 2003, 2007, 2008; DeCelles et al., 2014; Wu et al., 2014; Wang et al., 2017). The geologic evidence includes 1) the marked slowdown in convergence velocity during the period 60–44 Ma (Lee and Lawver, 1995), 2) deformation along the continental margin (Beck et al., 1995; Wang et al., 2017), and 3) provenance analysis demonstrating a switch from passive margin-derived sediment to the first arrival of Asia-derived clastic detritus at ~60 Ma (DeCelles et al., 2014; Wu et al., 2014; Hu et al., 2016).

Based on the latest sedimentary and geochronological dating results from the Sangdanlin and Gyangzes foreland basins, Wu et al. (2014) found that the first arrival of Asia-derived clastic sediments deposited on the northern Indian continental margin occurred at 60 Ma in both basins, which are now located near the Indus-Yarlung Zangbo suture zone. This timing is earlier than previously thought. Furthermore, no evidence supports the existence of an intra-oceanic arc within the Neo-Tethys Ocean. DeCelles et al. (2014) also conducted similar work and concluded that the marked change in sandstone compositions from Indian to Asian provenances occurred between 60 and 58.5 ± 0.6 Ma.

Summarizing the convergence rates of the northward-drifting Indian continent relative to the Asian continent, Lee and Lawver (1995) found that a significant slowdown in convergence velocity occurred at 60–44 Ma and concluded that this slowdown represents the initial contact between India and Asia. Their results are also consistent with our analysis based on the paleolatitudinal overlap between India and Asia (DeCelles et al., 2014; Wu et al., 2014; Hu et al., 2016).

Due to their proximity to the India–Asia suture zone, the strata along the southernmost margin of the Lhasa block have experienced intense deformation and magmatic activity, and their paleomagnetic information may have been altered. Therefore, this paleomagnetic and geochronological study was conducted far from the southernmost margin in order to avoid such influences. However, our results are still in full agreement with certain robust results from the southern margin of the block. For instance, data obtained in this study are compatible with the results of Tang et al. (2013), Achache et al. (1984), and Chen et al. (2010) at the 95% probability level, which further validates our results.

However, as proposed and summarized by Lippert et al. (2014), paleomagnetically determined ages usually represent a minimum estimate. Geologically, a forearc must have separated the Gangdese arc from the subduction zone prior to collision, and relics of a Cretaceous forearc are probably preserved in the IYZSZ (Einsele et al., 1994; Yin et al., 1994). The typical arc-trench width today is approximately 200–300 km (Lippert et al., 2014). Therefore, another 2° of distance should be added south of the modern southern boundary of the Lhasa block (*i.e.*, at 11.1 ± 2.7°). Thus, the age of initial collision between the Tethyan–Himalaya and southern Tibet should be even earlier than the age proposed here, likely ~61–62 Ma. No direct geologic evidence for latitudinal shortening estimates within the suture zone have been available until now; hence, the conservative conclusion is that the collision initiated at 59.3 Ma at 13.1 ± 2.7°N.

7.3. Latitudinal shortening estimates for Tibet and stable Asia

Our results from the northern part of the Lhasa block provide a key constraint on precisely reconstructing the paleogeographic position of the Lhasa block during the Early Cretaceous. The outcrops in the northern part of the Lhasa block experienced much less intense magmatic activity and deformation than those from the southernmost margin of the Lhasa block. By quantitatively comparing paleolatitudinal differences in the southernmost margin of Asia (*i.e.*, the Lhasa block) and the northernmost margin of Greater India, the overlap between the two paleolatitudes with a certain confidence level represents the onset of Indo-Asia collision, as discussed above (Section 7.2).

Based on abundant geological evidence, including ophiolitic mélange dating, geochemical research, and structural geological analysis, from both sides of the Bangong–Nujiang suture zone (northern margin of the Lhasa block and southern margin of the Qiangtang block), the closure of the Bangong–Nujiang Ocean was likely completed by the Late Jurassic–Early Cretaceous (Dewey et al., 1988; Chen et al., 1993b; Murphy et al., 2000; Kapp et al., 2003, 2005, 2007; Chu et al., 2006; Guynn et al., 2006; Zhang et al., 2011). Therefore, the Lhasa block represented the southernmost margin of Asia between the Early Cretaceous (~140–130 Ma) and the India–Asia collision, and the blocks that comprise Tibet kinematically acted as a unified entity during this time (Dewey et al., 1988; Chen et al., 1993b; Metcalfe, 1996; Murphy et al., 2000; Kapp et al., 2003, 2005; Guynn et al., 2006; Metcalfe, 2006; DeCelles et al., 2007; Kapp et al., 2007; Zhang et al., 2011). The relationship between the Lhasa block to the south and the Qiangtang block to the north evolved from oceanic subduction to intracontinental contraction (Dewey et al., 1988; Chen et al., 1993b; Metcalfe, 1996; Murphy et al., 2000; Kapp et al., 2003, 2005; Guynn et al., 2006; Metcalfe, 2006; DeCelles et al., 2007; Kapp et al., 2007; Zhang et al., 2011).

Paleomagnetic studies have confirmed that the Lhasa block remained essentially stationary between the Early Cretaceous and the early Paleogene prior to the India–Asia collision, with little N–S movement occurring during this time interval (Achache et al., 1984; Sun et al., 2008; Chen et al., 2010; Sun et al., 2010; Chen et al., 2012; Meng et al., 2012; Sun et al., 2012; Ma et al., 2014; Yang et al., 2015). The sub-plots of Fig. 8 (a and b) display the equal-area projections of the paleopoles from this study, previously published results from the Lhasa and Tethyan–Himalayan blocks, and the APWP reference poles for the Eurasian continent and Indian shield. The latitudinal difference, Δλ, between the Lhasa block and the 120 Ma Eurasian continent reference pole is 13.2 ± 3.7° for the Early Cretaceous (Fig. 8a). Therefore, 1450 ± 400 km of crustal shortening has occurred across Tibet and stable Asia since that time. The same analysis was applied to the lower Cenozoic results (Fig. 8b; Table 2). The Δλ between the Lhasa block and the 60 Ma Eurasian continent reference pole is 15.8 ± 6.1° for the early Cenozoic, which is indicative of 1740 ± 670 km of convergence across Tibet and stable Asia.

The amount of convergence has long been a highly controversial subject, despite of and in part because of its importance to constraining the orogenesis, plateau surface uplift, and evolution of the Himalayas and the Tibetan Plateau. This issue is closely related to determining the age of the onset of the India–Asia collision. Thus, estimates vary greatly between different scholars in the Earth science community. The researchers that argue for relatively small magnitudes of latitudinal shortening across southern Tibet and stable Asia also argue for a much later India–Asia collision age within a range of 55–34 Ma (Aitchison and Ali, 2001; Aitchison et al., 2007; Ali and Aitchison, 2008; Tan et al., 2010; Chen et al., 2012; Meng et al., 2012; van Hinsbergen et al., 2012). For example, Tan et al. (2010) proposed relatively low crustal shortening amounts of 1.7 ± 2.6° (188 ± 280 km) to 5.6 ± 5.4° (620 ± 600 km) based on the Upper Cretaceous Shexing Formation red beds (100–70 Ma) and Upper Cretaceous lava flows. The recent results of Meng et al. (2012) tightly constrain the timing of India–Asia collision and are consistent (<600 km) with the results of Tan et al. (2010). Although these results passed the fold test and certain data points were inclination-corrected, the data are not corroborated by abundant geological and geophysical evidence (Dewey et al., 1988; Searle, 1988; Dewey et al., 1989; Le Fort, 1989; Yin and Harrison, 2000; Johnson, 2002; Mo et al., 2003; Ding et al., 2005; Mo et al., 2007, 2008; Cai et al., 2011; Gibbons et al., 2015; Hu et al., 2016; Jiang et al., 2016; Wang et al., 2017). Thus, more detailed and systematic paleomagnetic works are still needed to support these estimates.

Our latitudinal shortening estimate is consistent with certain important paleomagnetic works (Achache et al., 1984; Besse et al., 1984; Chen et al., 2010; Sun et al., 2010, 2012; Tang et al., 2013). The recent results of Tang et al. (2013) (1900 ± 700 km of N–S convergence between the Lhasa block and southern Siberia), Ma et al. (2014) (1000 ± 300 km), and Yang et al. (2015) (780 ± 240 km) also vary widely. The total crustal shortening estimates all come from magnetic anomalies, paleomagnetic works, and volumetric balancing estimates (Achache et al., 1984; Besse et al., 1984; Patriat and Achache, 1984; Dewey et al., 1988; Ratschbacher et al. 1994; Johnson, 2002), but more works are needed.

Since the onset of the India-Asia collision, the most significant factor in the northward latitudinal convergence between the Lhasa block and stable Asia has been the continuing India–Asia collisional event (Molnar and Tapponnier, 1975; Patriat and Achache, 1984; Yin and Harrison, 2000; Besse and Courtillot, 2002). The amount of N–S latitudinal shortening suggested by the paleomagnetic data could be accounted for by geological observations in the form of internal deformation, such as a

series of fold–thrust belts (*e.g.*, the Shiquanhe–Gaize–Amdo thrust system); normal faults (in central Tibet); large lateral displacements (tectonic extrusion) along strike–slip faults (*e.g.*, the Kunlun, Altyn–Tagh, and Red River strike–slip faults); strike–slip faulting along the Karakorum fault; and crustal thickening in the southern margin of the Lhasa block, the Qiangtang block, the Kunlun Shan, Qilian Shan, Tianshan, Altai, and other mountain ranges. Additionally, intracontinental subduction beneath Qaidam and Tibet and the southeasterly continental extrusion of the Indo-China block could also accommodate a share of the shortening (Molnar and Tapponnier, 1975; Tapponnier et al., 1982; Coward et al., 1988; Royden et al., 1997; Meyer et al., 1998; Yin and Harrison, 2000; Yin, 2006; Lippert et al., 2011).

8. Conclusions

New paleomagnetic and geochronological data from the Duoni Formation in the northern part of the Lhasa block provide accurate estimates of the paleogeographic position of the Lhasa block, the timing of the onset of the India–Asia collision and the corresponding amount of latitudinal continental shortening since the Early Cretaceous (~120 Ma).

The geochronological dating of the Duoni Formation volcanics provides a robust result of 120.2 ± 0.5 Ma, consistent with previously published results. Based on the geochronological data, a high-temperature characteristic magnetic component was successfully isolated and passed the fold and stepwise unfolding tests, permitting us to interpret these data as primary thermal remanent magnetizations for the volcanics and primary detrital remanence for the sandstones. Combining our obtained paleopole (derived from 19 site-mean directions) with previously published poles, the paleolatitudes 13.1 ± 2.7° (Early Cretaceous) and 16.3 ± 5.6° (early Cenozoic) were obtained for the southernmost margin of the Lhasa block (reference site: 30.0°N, 91.1°E). Between the Early Cretaceous (120 Ma) and the initiation of India–Asia collision in the Early Paleogene (59.3 Ma), the southernmost margin of the Lhasa block remained essentially stationary, and no significant latitudinal motion occurred between southern Tibet and stable Asia.

Relative to the 120 Ma and 60 Ma APW reference poles for Eurasia, our results indicate that 1450 ± 400 km and 1740 ± 67 km of latitudinal shortening, respectively, is estimated to have occurred between southern Tibet and stable Eurasia. In summary, since the initiation of India–Asia collision, at least 1500 km of latitudinal shortening has occurred across Tibet and stable Asia. This magnitude of N–S crustal shortening was accommodated by internal deformation, normal faults, large lateral displacements along strike–slip faults, crustal thickening in the composite blocks and mountain ranges, intracontinental subduction beneath Qaidam and Tibet, and southeasterly continental extrusion of the Indo-China block.

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